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Supplemental Material

Bayesian Estimation of Human Population Toxicokinetics of PFOA, PFOS, PFHxS, and PFNA from Studies of Contaminated Drinking Water

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Figure S2. Perfluorooctanoic acid (PFOS) serum concentrations as data (points) and predictions (boxplots) for studies other than Decatur, AL. These include Minnesota (USA Lake Elmo/ Cottage Grove), Horsham (USA PA), Paulsboro (USA NJ), Warminster (USA PA), and Warrington (USA PA). Shading indicates that samples were all taken at a single time (T in years) point. This figure shows that the training data points generally fall within the interquartile range of the predicted serum concentrations. For testing data, most serum measurements (points) fall within the interquartile range, and few are outside of the upper and lower bounds of estimates. More detailed tabular data can be found in the project repository <https://github.com/wachiuphd/2022-Bayes-PFAS-PK>.

Figure S3. Perfluorononanoic acid (PFNA) data (points) and predictions (boxplots) for studies other than Decatur, AL. PFNA summary data included Paulsboro (USA NJ), Horsham (USA PA), and Warminster (USA PA). Shading indicates that samples were all taken at a single time (T in years) point. This figure shows that the all measured serum concentrations (points) fall within the interquartile range of the predicted serum for both training and testing data. More detailed tabular data can be found in the project repository <https://github.com/wachiuphd/2022-Bayes-PFAS-PK>.

Figure S4. Perfluorohexane sulfonate (PFHxS) data (filled circles) and predictions (boxplots) for studies other than Decatur, AL. PFHxS summary data studies included Paulsboro (USA NJ), Horsham (USA PA), Warminster (USA PA), and Warrington (USA PA). Shading indicates that samples were all taken at a single time (T in years) point. This figure shows that the measured serum concentrations (points) fall within the range of the predicted serum concentrations for both training and testing data. More detailed tabular data can be found in the project repository <https://github.com/wachiuphd/2022-Bayes-PFAS-PK>.

Figure S5. Comparison of V_d (left) and k (right) estimates across individuals from different datasets using the primary model. Specifically, for each individual a random estimate was drawn from the MCMC samples for their z-score for the corresponding parameter, and the resulting histogram compared to the expected standard normal distribution. Perfluorononanoic acid (PFNA) and perfluorohexane sulfonate (PFHxS) included only one individual-level dataset, so only comparisons for perfluorooctanoic acid (PFOA: top) and perfluorooctane sulfonic acid (PFOS: bottom) are included. There is no systematic effect of dataset origin (country or city) on V_d or k , and individual samples from each city were consistent with the expected standard normal distribution by one-sample Kolmogorov-Smirnov test (two-sided $p > 0.05$). More detailed information can be found in the project repository <https://github.com/wachiuphd/2022-Bayes-PFAS-PK>.

Figure S6. Results of sensitivity analysis showing posterior estimates of geometric mean of half-life (dotted line) for each of 9 alternative model definitions (described in methods and in Table S3). all-train = Use all data for training, Cbgd-80 = Primary analysis with background relative source contribution = 20%, Cbgd-100 = Background relative source contribution = 0%, Cbgd-20 = Background relative source contribution = 80%, Vd-d20 = Volume of distribution prior decreased 20%, Vd-i20 = Volume of distribution prior increased 20%, indiv = Use only individual data, pop = Use only population data, swap = Swap training and testing data, PFOA = perfluorooctanoic acid, PFOS = perfluorooctane sulfonic acid, PFNA = perfluorononanoic acid, PFHXS = perfluorohexane sulfonate. . More detailed information can be found in the project repository <https://github.com/wachiuphd/2022-Bayes-PFAS-PK>.

Figure S7. Results of sensitivity analysis showing posterior estimates of geometric standard deviation (GSD) of half-life (dotted line) for each of 9 alternative model definitions (described in methods and in Table S3). all-train = Use all data for training, Cbgd-80 = Primary analysis with background relative source contribution = 20%, Cbgd-100 = Background relative source contribution = 0%, Cbgd-20 = Background relative source contribution = 80%, Vd-d20 = Volume of distribution prior decreased 20%, Vd-i20 = Volume of distribution prior increased 20%, indiv = Use only individual data, pop = Use only population data, swap = Swap training and testing data, PFOA = perfluorooctanoic acid, PFOS = perfluorooctane sulfonic acid, PFNA = perfluorononanoic acid, PFHXS = perfluorohexane sulfonate. More detailed information can be found in the project repository <https://github.com/wachiuphd/2022-Bayes-PFAS-PK>.

Figure S8. Results of sensitivity analysis showing posterior estimates of geometric mean (GM) of Volume of distribution (V_d) for 9 different model formulations. all-train = Use all data for training, Cbgd-80 = Primary analysis with background relative source contribution = 20%, Cbgd-100 = Background relative source contribution = 0%, Cbgd-20 = Background relative source contribution = 80%, Vd-d20 = Volume of distribution prior decreased 20%, Vd-i20 = Volume of distribution prior increased 20%, indiv = Use only individual data, pop = Use only population data, swap = Swap training and testing data, PFOA = perfluorooctanoic acid, PFOS = perfluorooctane sulfonic acid, PFNA = perfluorononanoic acid, PFHXS = perfluorohexane sulfonate. More detailed information can be found in the project repository <https://github.com/wachiuphd/2022-Bayes-PFAS-PK>.

Figure S9. Results of sensitivity analysis for GSD of Vd. Cbgd-80 represents the primary analysis. all-train = Use all data for training, Cbgd-80 = Primary analysis with background relative source contribution = 20%, Cbgd-100 = Background relative source contribution = 0%, Cbgd-20 = Background relative source contribution = 80%, Vd-d20 = Volume of distribution prior decreased 20%, Vd-i20 = Volume of distribution prior increased 20%, indiv = Use only individual data, pop = Use only population data, swap = Swap training and testing data, PFOA = perfluorooctanoic acid, PFOS = perfluorooctane sulfonic acid, PFNA = perfluorononanoic acid, PFHXS = perfluorohexane sulfonate. More detailed information can be found in the project repository <https://github.com/wachiuphd/2022-Bayes-PFAS-PK>.

MCSim Model Code

References

Table S1. Detailed study descriptions; description of data sources for water and serum concentrations by perfluoroalkyl substance (PFAS) and location.

Location	Source	Data extracted	Study description
PFOA			
Arnsberg, Germany – data for only Huesten and Neheim boroughs	Skutlarek et al. 2006	PFOA in Water	Note: This study is included for reference and dates of known contamination and intervention. Water concentrations from Holzer et al., 2009 were used for parameter estimation. During several campaigns in 2006, surface and drinking water samples were collected from rivers, canals, and public buildings from Rune in March, and from the Ruhr area and Moehne in May. The highest values were found at Neheim where the drinking water originates from the Moehne River; the major component was PFOA at 0.519 µg/L. Hölzer et al. (2008) provided paired drinking water concentrations and was used in the Bayesian modeling process.
Arnsberg, Germany (defined by 4 boroughs: Neheim, Huesten, Herdringen, and Bruchhausen)	Hölzer et al. 2008	PFOA in Serum and water	Note: This study is included for reference and dates of known contamination and intervention. Water concentrations from Holzer et al., 2009 were used for parameter estimation. Between 2000 and 2006, soil conditioner mixed with industrial waste had been applied to 800 agricultural sites along the Moehne and Upper Ruhr rivers. This soil conditioner was determined to be the main source of PFAS contamination. In May 2006, residents in parts of Arnsberg were notified by local health authorities to not use their drinking water to prepare baby food, and pregnant women were advised to avoid regular intake of the water. Subsequently, over half of mother-child pairs and over a quarter of the men in Arnsberg voluntarily reduced their consumption of the drinking water. Data on PFAS levels in drinking water in Arnsberg are available starting in May 2006. As reported by Skutlarek et al. (2006), PFAS levels in May 2006 were 0.519 µg/L. The PFOA-contaminated water supplied to Arnsberg residents was from the Moehnebogen waterworks. In response to the high levels of PFAS detected, activated charcoal filters were installed in 2006 by the waterworks. The study recruited 355 participants (101 males, 164 females, and 90 children) who lived in the Arnsberg area. Study blood and water sample collection occurred from October 11 to November 8, 2006. On the same day that blood samples were drawn, participants were directed to collect their own tap water samples from their kitchens using written instructions given by the study staff. However, the study only provides a range of PFAS water levels over the month that blood samples were drawn. During the study period, the PFOA concentrations in tap water samples increased from below the levels of detection (LOD) to 0.071 µg/L due to diminished filtration performance.

Location	Source	Data extracted	Study description
Arnsberg, Germany	Hölzer et al. 2009	PFOA in Serum and water	This study recruited participants who had participated in Hölzer et al. (2008), recruiting and analyzing blood samples from 288 participants (138 mothers, 82 men, 68 children). Samples included initial blood sampling and follow-up sampling collected 335 to 386 days apart (median: 359 days) in 2007. From 2006 to 2007, the geometric mean PFOA concentrations in the blood plasma of Arnsberg residents decreased from 22.1 to 17.4 µg/L in children, from 23.8 to 18.8 µg/L in mothers, and from 25.7 to 23.4 µg/L in men.
Horsham, Pennsylvania, USA	Pennsylvania Department of Health 2019	PFOA in Serum	The PFAS Exposure Assessment Technical Toolkit (PEAT) Pilot Project randomly selected 235 community members who lived in areas served by four water systems: Horsham Water and Sewer Authority, Warminster Municipal Authority, Warrington Township Water and Sewer, and North Wales Water Authority. Serum samples from community members were collected and analyzed for 11 PFAS compounds during May through September 2018. Community members were considered eligible if they lived in one of four service areas prior to July 2016. After this date, all public wells that exceeded U.S. Environmental Protection Agency (EPA) lifetime health advisory limit (LHAL) levels for PFOS were taken out of service; residents on private wells were switched to bottled water. Study participants were recruited in May 2018, and blood serum was collected from May through September 2018. Of the 235 participants, 12 were aged 3–11 years, 19 were aged 12–19 years, and 204 were aged 20 years or older. Additionally, more than half of the participants (55.7%) were females, and 193 participants were public water consumers. The mean serum PFOA level of public water drinkers serviced by the Horsham Water and Sewer Authority (n = 61) was 3.65 µg/L [95% Confidence Interval (CI): 2.89–4.60] in May–September 2018.
Horsham, Pennsylvania, USA	EPA 2017	PFOA in Water	PFOA water concentration data were collected from June 24, 2014 to March 19, 2015 for the ZIP code 19044 served by the Horsham Water and Sewer Authority. Thirty-four samples were taken, of which six exceeded the reporting limit. The geometric mean of samples was 0.042 µg/L, the geometric standard deviation (GSD) was 2.79, the median was 0.026 µg/L, and the arithmetic mean was 0.075 µg/L.
Horsham, Pennsylvania, USA	HWSA 2014, 2018	PFOA in Water	Note: This study is included for reference and dates of known contamination and intervention. The EPA 2017 finished water concentrations were used for parameter estimation. The Horsham Water and Sewer Authority 2014 Water Quality Report provides PFOA and PFOS results for water samples taken from June to December 2014. The average PFOA concentration in all Horsham public water wells was below the reporting limit of 0.3 µg/L. The report states that two wells were found to exceed federal guidelines for PFOS concentrations and were shut off, but PFOA concentrations were not of concern. Contaminated wells were taken out of service in 2016. The Horsham Water and Sewer Authority 2018 Water Quality Report provides PFOA and PFOS sample results for seven wells that provided drinking water in 2018. Dates and the number of samples taken are not provided, and two wells were out of service for part of the year.

Location	Source	Data extracted	Study description
Lake Elmo/ Cottage Grove, Minnesota, USA	Johnson et al. 2017	PFOA in Water and serum	Groundwater in Washington County, Minnesota, was contaminated with PFHxS from a nearby waste disposal facility in 2004. In August 2008, surveys were mailed to 500 households that were randomly selected from Oakdale municipal billing records and to 169 households served by private wells contaminated with ≥ 0.1 $\mu\text{g/L}$ of PFOA or PFOS. The study enrolled 98 participants served by municipal water and 98 participants served by private well water. Participant blood serum were measured for PFOA, PFOS, and PFHxS, and collection was completed by December 31, 2008. PFOA and PFOS concentrations in drinking water were available only for private well users, as samples had been collected in these homes from 2005 to 2008. Individual-level data were provided graphically for the correlation of PFOA and PFOS water concentration with serum PFOA or PFOS concentration for each individual. For the private well community, the geometric mean PFOA serum concentration was 13.6 $\mu\text{g/L}$ (median: 13 $\mu\text{g/L}$, range: 1.6–177 $\mu\text{g/L}$) for serum collected from August 1 to December 31, 2008. For drinking water collected from 2005 to 2008, the average PFOA concentration was 0.04–1.87 $\mu\text{g/L}$.

Location	Source	Data extracted	Study description
Little Hocking, Ohio, USA	Bartell et al. 2010	PFOA in Water and serum	The C8 Health Project was initiated from a class action settlement agreement filed against E.I. du Pont de Nemours and Company, which appointed the C8 Science Panel to assess a link between PFOA exposure and diseases in several affected mid-Ohio Valley communities. The study population consisted of adults (18–89 years of age) who were recruited from the C8 Health Project and primarily consumed public drinking water between 2005 and 2007 in Lubeck (n = 132) or Little Hocking (n = 40). Additionally, 17 bottled water drinkers were recruited from Lubeck and 11 from Little Hocking. 75% of the participants recruited were from Lubeck since water filtration began shortly after baseline. The remaining 25% of participants were recruited from Little Hocking, where filtration began five months after baseline interviews, and served as a comparison group. An approximately equal number of males and females were recruited. Participant eligibility criteria included (1) serum PFOA concentrations ≥ 50 $\mu\text{g/L}$, (2) residential water service provided by the Lubeck Public Service District or the Little Hocking Water Association, and (3) participants did not grow vegetables at home. Public water participants had to report consumption at home as their primary use of public water in 2005 to 2007, and report that public water was used for cooking, showering, and bathing in 2007. Bottled water drinkers had to report that bottled water was their primary source of residential water consumption in 2005, 2006, and 2007. Participants were enrolled, and baseline blood samples were taken in May–June 2007. Baseline serum PFOA concentration in Little Hocking for public water drinkers was 424 $\mu\text{g/L}$ (± 333 $\mu\text{g/L}$) and for bottled water drinkers was 180 $\mu\text{g/L}$ (± 193 $\mu\text{g/L}$). Pre-filtration water PFOA concentrations in Little Hocking ranged from 1.9 to 4.9 $\mu\text{g/L}$, and post-filtration water PFOA concentrations were all non-quantifiable (< 0.016 $\mu\text{g/L}$) or NDs. The results are presented as the percentage change in mean individual serum PFOA concentrations from baseline to 12 months after baseline. For public water drinkers, PFOA concentrations decreased by 23%, and for bottled water drinkers by 27%. From November–December 2007 to August 2008, mean serum concentrations decreased by 15% for public water drinkers and 14% for bottled water drinkers. Data caveats: After the 6-month blood collection, 10% of public water drinkers reported that they were now primarily drinking bottled water. After the 12-month blood collection, 39% of bottled water drinkers reported that they were now primarily drinking public water.

Little Hocking, Ohio, USA	Emmett et al. 2006	PFOA in Water and serum	PFOA was detected in the Little Hocking water system in January 2002 to May 2005 at average concentrations of 3.55 µg/L (range: 1.5 to 7.2 µg/L). For study participants using private wells, PFOA concentrations ranged from ND to 14 µg/L. The study recruited two populations: residents potentially exposed to PFOA in both air ($\geq 0.2 \mu\text{g}/\text{m}^3$) and water, and residents potentially exposed to PFOA in water with minimal exposure in air. All randomly selected participants resided in the Little Hocking water systems for two years as of July 2004, were ≥ 2 years of age, and had no bleeding disorders. A control group from Philadelphia was recruited in which there were 324 adult and child participants from 161 households in the study. Additionally, 18 individuals from the study were classified as the “substantial occupational exposure” group. Median serum PFOA concentrations for this group (775 µg/L) were significantly higher than individuals with no occupational exposure (329 µg/L). Separately, 48 individuals reported past or current potential PFOA exposure but there were no significant differences in serum PFOA levels (388 µg/L) compared to controls. These 48 individuals were categorized into a “potential exposure” group for statistical analysis. Substantial occupational exposure individuals were excluded from the community analysis. Blood samples were collected between July 2004 to February 2005. There were 291 adults and children who resided in areas serviced by the Little Hocking Water Association for at least two years. The median PFOA serum concentration for this group was 374 µg/L [mean: 448, range 7–1,950, interquartile range (IQR): 221–576, units: µg/L]. Comparatively, 26 participants serviced by public water and bottled or spring water had median serum concentrations of 320 µg/L (mean: 358, range: 72–1,280, IQR: 206–370, units: µg/L). For the 10 participants who used bottled, cistern, or spring water as their primary source, median serum concentrations were 71 µg/L (mean: 154, range: 12–527, IQR: 49–217, units: µg/L). Overall, for all residents served by the Little Hocking water service district, median serum PFOA concentrations ranged from 298 to 370 µg/L. PFOA concentrations through air pollution did not result in a statistically significant difference of serum concentrations. No significant associations were found between serum PFOA levels and smoking, alcohol consumption, years lived at current residence, or gender. Participants > 60 years of age had significantly higher serum PFOA concentrations compared to all other groups except children 2–5 years old, indicating these may be particularly vulnerable groups. Tap water drinks per day, fruit and vegetables servings per week from a local garden, and use of a carbon filter were associated with higher-serum PFOA concentrations. The mean PFOA concentration in the Little Hocking water system in January 2002–May 2005 was 3.55 µg/L (minimum: 1.5 µg/L, maximum: 7.2 µg/L). Median serum levels for 291 adults and children (≥ 4 years of age) was 374 µg/L (mean: 448, range 7–1,950, IQR: 221–576) measured from July 2004 to February 2005. Mean and medium serum PFOA concentrations were provided per age group graphically. These were estimated for the following adult age groups: aged 16–20 years, mean: 310 µg/L, median: 250 µg/L; aged 21–30 years, mean: 250 µg/L, median: 240 µg/L; aged 31–40 years, mean: 320 µg/L, median: 310 µg/L; aged 41–50 years, mean: 350 µg/L, median: 310 µg/L; aged 51–60 years, mean: 410 µg/L, median: 240 µg/L; and aged 60+ years, mean: 600 µg/L, median: 475 µg/L.
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Location	Source	Data extracted	Study description
Lubeck, Ohio, USA	Bartell et al. 2010	PFOA in Water and serum	Study methods and descriptions of participants for this study is described above. In Lubeck, mean PFOA in drinking pre-filtration water ranged from 0.41 to 1.0 µg/L in samples taken from 2001 to 2006. In post-filtration water, PFOS was less than quantifiable (< 0.016 µg/L). Mean serum PFOA concentration in 132 public water drinkers in May–June 2007 was 122 µg/L (± 81 µg/L) and for bottled water drinkers was 58 µg/L (± 36 µg/L). Mean serum concentrations were measured 12 months after baseline in August 2008 and decreased by 26% for public water drinkers and by 27% for bottled-water drinkers. From November–December 2007 to August 2008, mean serum concentrations for public-water drinkers decreased by 15% and for bottled-water drinkers decreased by 14%. Serum PFOA concentrations for participants in the Lubeck Public Water District for people on public water supply declined from 122 µg/L [standard deviation (SD) = 81.0] in May–June 2007 to 103.82 µg/L (SD = 80.92) to 90.28 µg/L (SD = 80.91) in August 2008. For bottled-water drinkers in the same time periods, serum PFOA concentrations declined less precipitously from 58 µg/L (SD = 36) to 48.69 µg/L (SD = 32.87) to 42.34 µg/L (SD = 39.60).
North Wales, Pennsylvania, USA	Pennsylvania Department of Health 2019	PFOA in Serum	Serum levels for North Wales are reported in the PEATT Pilot Project, and the methodology and study description are described in further detail for Horsham, PA, above. Of note, there were 18 consumers of public water being serviced by both the Warrington Township Water and the Sewer and North Wales Water Authority. The mean serum PFOA level of public water drinkers serviced by the Warrington Township and the North Wales Water Authority (n = 18) was 1.63 µg/L (95% CI: 1.25–2.11) in May–September 2018.
North Wales, Pennsylvania, USA	NWWA 2018	PFOA in Water	Note: Due to uncertainties in mapping the correct water concentrations to this subset of consumers, the serum and water data for North Wales were not included in the analysis. In 2019, the North Wales Water Authority completed its acquisition of the Warrington Township Water and Sewer. As stated below, Warrington reported a PFOA concentration of 0.004 µg/L (range: 0.003–0.045 µg/L) in its 2018 water quality report. The North Wales Water Authority's 2018 Water Quality Report reports a PFOA concentration of 0.004 µg/L (range: 0.0039–0.0045 µg/L) in 2018. North Wales did not include PFAS levels in its water system quality report before 2018. The same water concentration was reported in both water supplies; therefore, it may be reasonable to assume that households serviced by both water systems were likely exposed to 0.004 µg/L (range: 0.0039–0.0045 µg/L) of PFOA in their drinking water in 2018

Location	Source	Data extracted	Study description
Paulsboro , New Jersey, USA	Graber et al. 2019	PFOA in Water and serum	In 2011 and 2012, the New Jersey Department of Environmental Protection monitored PFAS in raw and finished drinking water for several communities, including Paulsboro, New Jersey. A 2016 class-action lawsuit allowed Paulsboro, NJ residents to have one blood serum test for 13 PFAS. This cross-sectional study recruited residents who had claimed this blood test. Residents were also required to have lived in the water service area for at least 12 months prior to April 19, 2014, when remediation of the community water supply began. Recruitment occurred from November 14, 2016, to January 21, 2017, and required eligible adults to hand in their own and/or their children's/wards' PFAS blood results. Adults completed surveys of demographic information and health outcomes. Of note, the study team did not take part in the blood testing, as they only received testing done as a result of the class-action settlement. The study enrolled 186 Paulsboro residents, 165 of whom were over the age of 12 years. The median age was 47 years old. The geometric mean PFOA serum concentration for these 165 residents was 3.03 µg/L (95% CI: 2.70, 3.40) in 2016. In 2011-2012, PFOA concentrations in Paulsboro community water was 0.026 µg/L (also found in Post et al., 2013).

Location	Source	Data extracted	Study description
Ronneby, Sweden	Li et al. 2018	PFOA in Water and serum	Note: This study was used to inform the half-life priors. However, summary statistics for serum concentrations reported in the manuscript combined samples taken from pre- and post- intervention time periods, so these were not included in training or testing datasets. Outgoing drinking from two municipal waterworks (Brantafor and Karragarden) in Ronneby, Sweden were tested for PFASs on December 10, 2013. One third of households were serviced by Brantafor, which had an outgoing drinking water PFOA concentration of 0.1 µg/L. All other households were serviced by Karragarden, which had outgoing water PFOA concentrations of 0.001 µg/L. Brantafor was closed on December 16, 2013 and serviced households were switched to Karragarden for water, and a few days later the distribution network had no elevated PFAS levels. In May 2014, Brantafor waterworks was reopened for a trial period until October 2014. The sum of all PFASs in water was at most 0.04 µg/L during this time period. Levels for each type of PFAS was not provided. In June 2014, approximately 6 months after exposure to Brantafor contaminated water had ended, biomonitoring of the population was started. A panel study group (n=106) was formed in June 2014 consisting of individuals exposed to Brantafor drinking water. Ages ranged from 4 to 83 years, and 53% were women. Of the participants, 20 were men aged 15-50, and 30 were menstruating women. Samples for serum PFOS, PFOA, and PFHxS levels were collected baseline in June 2014 and on average 6 more times until September 2016. In June 2014, mean serum PFOA concentration among all participants was 21.1 µg/L ± 14.7 (min: 2.38, median: 17.5, max: 92). Mean half-life for PFOA among all participants was 2.7 years, among men aged 15-50 years was 2.8 years, and among women aged 15-50 years was 2.4 years. There was a strong age-related effect with 1.1% increase in PFOA serum concentrations per year of age. Gender and BMI were not associated with serum levels. At testing in September 2016, the mean PFOA serum decreased by 38% compared to June 2014. Water and serum concentrations for this panel study in Brantafor are summarized below: Water PFOA concentrations in Brantafor, Ronneby taken on December 10, 2013, averaged 100 µg/L. Corresponding serum PFOA taken in June 2014 had mean(± sd) of 21.1 (± 14.7) µg/L with a range of 2.38 - 92 and a median of 17.5 µg/L (n=106). A “Main Ronneby” group of 3418 residents was recruited from 2014 to 2016 from the entire Ronneby municipality. Residents exposed to the contaminated and non-contaminated waterworks were recruited to ensure broad serum concentrations. Mean serum concentrations of PFOA at baseline were 13.7 µg/L ± 12.0 (min: <0.4, median: 10.4, max: 91.9). A reference group of 242 residents from an unexposed municipality (Karlshamn) was examined in 2016 and had mean serum PFOA concentrations of 1.77 µg/L ± 0.81 (min: <0.4, median: 1.59, max: 4.98). The Main Ronneby group had median serum PFOA concentrations 6 times higher than the reference group.

Location	Source	Data extracted	Study description
Uppsala County, Sweden	Gyllenhammar et al. 2019	PFOA in Water and serum	Note: These data were not used in the analysis because time of cumulative drinking water exposure was reported, rather than either an average concentration of drinking water or concentrations on specific dates. Drinking water in Uppsala had been contaminated with PFAS for at least 20 years until 2012 when the contaminated wells were either closed or their use was severely restricted. This study investigated the determinants of PFAS serum concentrations in children ages 4, 8, and 12 years by focusing on maternal PFAS concentration at delivery, nursing history, and child drinking water exposure. Pregnant mothers were recruited during 1996 to 1999 or shortly after delivery in 2000 to 2011 and given lifestyle factor questionnaires. Researchers followed up with the children of these mothers were followed up with at ages 4, 8, and 12 years (2008 to 2015); however, no child was sampled at all three ages. Nine tap water samples were collected in Uppsala County in 2012 when contamination was first discovered. The exposure for Uppsala residents based on their location was estimated for 1996 to 2012 by modeling the distribution of contaminated well water in these years.
Warminster, Pennsylvania, USA	Pennsylvania Department of Health 2019	PFOA in Serum	Serum levels for Warminster, PA are reported in the PEATT Pilot Project, and the methodology and study description are provided in further detail for Horsham, PA above. Among public drinking water consumers serviced by the Warminster Municipal Authority (n=83), the mean serum PFOA concentration was 3.24 µg/L (95% CI: 2.73-3.84) between May and September 2018.
Warminster, Pennsylvania, USA	EPA 2017	PFOA in Water	PFOA water concentration data were reported for the zip code 18974 – Warminster Municipal Authority from November 19, 2013 to 8/19/2014, with a total of 38 samples, 13 of which had concentrations above the reporting limit (0.02 µg/L). Samples showed an arithmetic mean of 0.091 µg/L, median of 0.034 µg/L, geometric mean 0.053 µg/L, and GSD of 2.79.
Warminster, Pennsylvania, USA	WMA 2017, 2018	PFOA in Water	The Warminster Municipal Water and Sewer Services' 2017 and 2018 water quality reports present the combined PFOA and PFOS concentrations in water. Water quality reports for previous years are not available. The combined PFOS+PFOA levels in Warminster wells in both 2017 and 2018 were non-detectable.
Warrington, Pennsylvania, USA	Pennsylvania Department of Health 2019	PFOA in Serum	Serum levels for Warrington, PA are reported in the PEATT Pilot Project, and the methodology and study description are provided in further detail for Horsham, PA above. Among public drinking water consumers serviced by only the Warrington Township Water and Sewer (n=31), the mean serum PFOA concentration was 3.63 µg/L (95% CI: 2.76-4.78) between May and September 2018.

Location	Source	Data extracted	Study description
Warrington, Pennsylvania, USA	Warrington Township 2014, 2018	PFOA in Water	The Warrington Township produces an annual water quality report of the public water system. During 2014, the western portion of Warrington was serviced by North Wales Water Authority, and the eastern portion was serviced by nine wells. The eastern and western systems are interconnected. The PFOA concentration in the Warrington Township water was 0.12 µg/L (range: 0 – 0.12 µg/L) from January 1 to December 31, 2014. During 2018, all water supplied by the Warrington Township was purchased from the North Wales Water Authority and Forest Park Water Treatment Plant. From January 1, 2018 to December 31, 2018, testing was conducted on the Western and Eastern systems of the Warrington Township water system. In 2018, the water PFOA level detected was 0.004 µg/L (range: 0.003 – 0.0045 µg/L). The source of contamination noted is firefighting foam and other man-made sources.
West Morgan/East Lawrence – Decatur, AL	ATSDR 2013	PFOA in Serum and water	In November 2008, EPA tested water samples from the West Morgan/East Lawrence (WM/EL) water system in Moulton, Decatur County in Alabama. The levels detected in the WM/EL system were below EPA's Provisional Health Advisory levels (PHAL) for PFOA (0.4 µg/L) and PFOS (0.2 µg/L). In February 2009, six private wells (PW) were tested for PFOA. One well had a concentration of 2.2 µg/L, another of 0.6 µg/L, and the other four were below the EPA Provisional level. Residents were given alternative drinking water and connected to the public water system. In September 2009, WM/EL system levels were below EPA's PHAL. From August 2009 to August 2010, EPA, Decatur Utilities, and local industry sampled 12 private drinking water wells. PFOS was detected at 0.365 µg/L at one location – the residence was provided a substitute source for drinking water and connected to their local public water supply. In 2011, WM/EL system were below EPA's PHAL. Between November 2008 and 2011, WM/EL levels can be presumed to be <0.4 µg/L (PFOA) and <0.2 µg/L (PFOS). Serum samples were collected from residents (n=121) who consumed public water from the WM/EL system. The geometric mean serum PFOA level was 17.59 µg/L (range: 2.2 to 78.8 µg/L and median of 18.1 µg/L) in blood collected from April 12-15, 2010. Geometric mean serum PFOA levels of all 153 participants including those with drinking water from other sources than WM/EL, was 16.3 µg/L (range: 2.2 to 144 µg/L).

Location	Source	Data extracted	Study description
West Morgan/East Lawrence – Decatur, AL	ATSDR 2016	PFOA in Serum and water	The investigation tested targeted populations in Morgan, Lawrence, and Limestone counties in Alabama. Individuals who had participated in a 2010 exposure investigation (153 residents) were targeted for blood and urine testing. The time period between 2016 and 2010 serum concentrations is 2095 days. Serum collection is estimated to occur on January 6, 2016. Blood samples from a total of 45 study participants were analyzed for eleven PFAS: PFOSA, Et- PFOSA-AcOH, Me-PFOSA-AcOH, PFHxS, PFNA, PFDeA, linear perfluorooctanoate (n-PFOA), sum of branched PFOA isomers (Sb-PFOA), linear perfluorooctane sulfonic acid (n-PFOS), sum of isomers of perfluorodimethylheptane sulfonic acid (Sm- PFOS), and sum of isomers of perfluorodimethylhexane sulfonic acid (Sm2-PFOS). On June 13, 2016, PFOA was measured in water below the detection limit. Serum PFOA concentrations declined from 16.3 µg/L (n = 153) in 2010 to 11.7 (µg/L) 2016 (n=45) GM, minimum, maximum, 95th% generated for: Total PFOA, total PFOS, PFNA, PFHxS, PFDeA, Me-PFOSA-AcOH, Et-PFOSA-AcOH, and PFOSA. PFAS concentrations below the limit of detection (LOD) were assigned a value equal to the LOD divided by the square root of two.
PFOS			
Arnsberg, Germany – relevant data for sites reported for only Huesten and Neheim	Skutlarek et al. 2006	PFOS in Water	Note: This site was not included in parameter estimates for PFOS due to lack of time-course drinking water data for PFOS. Given the fluctuation in the PFOA samples, we expected variability in PFOS as well, and this study was excluded due to low-quality water concentration data. The following data were reported specifically for PFOS: PFOS drinking water levels from Table 7 in study from Neheim, May 2006 were 0.005 µg/L. PFOS surface water levels from Table 6 in study were not detected in Heusten but were 0.014 µg/L for Neheim in May 2006. Further study details and descriptions are provided for this study under PFOA.
Arnsberg, Germany (defined by 4 boroughs: Neheim, Huesten, Herdringen and Bruchhausen)	Hölzer et al. 2008	PFOS in Serum and water	Note: This study is included for reference and dates of known contamination and intervention. This site was not included in parameter estimation for PFOS due to low-quality water concentration data. This study is described in more detail under PFOA. Serum PFOS levels (µg/L) in Arnsberg residents were sampled from October 11 to November 8. The geometric mean PFOS levels were 4.9 µg/L, 5.8 µg/L, and 10.5 µg/L; and standard deviations were 2.9 µg/L, 2.8 µg/L, and 6.1 µg/L in children, mothers, and men, respectively. The ranges of serum PFOS levels in children, mothers, and men were 2.4-20.6 µg/L, 1.7-16.7 µg/L, and 2.7-36.2 µg/L. Further study details and descriptions are provided for this study under PFOA.

Location	Source	Data extracted	Study description
Arnsberg, Germany (defined by 4 boroughs: Neheim, Huesten, Herdringen and Bruchhausen)	Hölzer et al. 2009	PFOS in Serum and water	Note: This study is included for reference and dates of known contamination and intervention. This site was not included in parameter estimation for PFOS due to low-quality water concentration data. Study details and descriptions are provided for this study under PFOA section of this table. The geometric mean PFOS-concentrations in blood plasma of Arnsberg residents from 2006 to 2007 decreased from 5 to 4.5 µg/L in children, from 5.8 to 5.3 µg/L in mothers, and from 10.6 to 10.2 µg/L in men within one year. From 2006 to 2007, the geometric mean of individual PFOS- concentrations decreased by 9% in children, 10% in mothers, and 8% in men.
Horsham, Pennsylvania	Pennsylvania Department of Health 2019	PFOS in Serum	Study description and details are provided for Horsham PA above under PFOA. The mean serum PFOS level of public water drinkers serviced by the Horsham Water and Sewer Authority (n=61) was 12.17 µg/L (95% CI: 9.03 – 16.39) in May–September 2018.
Horsham, Pennsylvania	EPA 2017	PFOS in Water	PFOS water concentration data was reported for the zip code 19044 – Horsham Water and Sewer Authority from June 24, 2014 to March 19, 2015 with a mean concentration of 0.2996 µg/L, a median of 0.097 µg/L, a geometric mean of 0.14 µg/L, and a GSD of 3.59. Thirty-four samples were taken, and seven were above the detection limit.
Horsham, Pennsylvania	HWSA 2014, 2018	PFOS in Water	Note: This study is included for reference and dates of known contamination and intervention. The EPA 2017 finished water concentrations were used for parameter estimation. The Horsham Water and Sewer Authority released a Water Quality Reports of its system in 2014 and 2018. The 2014 water quality reports present PFAS levels prior to contaminated wells being taken out of service in 2016. In water samples taken from June to December 2014, the average PFOS concentration in all Horsham public water wells was 0.06 µg/L (range: ND – 1.0 µg/L). The report does state that two wells were found to exceed federal guidelines for PFOS concentrations and were shut off, but PFOA concentrations were not of concern. In 2018, the mean PFOS concentration detected was 0.0075 µg/L (range: 0.0041 – 0.013 µg/L).
Lake Elmo/ Cottage Grove, Minnesota	Johnson et al. 2017	PFOS in Water and serum	Study details and participant recruitment are provided above under PFOA. For the private well community, the geometric mean PFOS serum concentration was 32.9 µg/L (median: 35 µg/L, range: 3.2–448 µg/L) for serum collected from August 1, 2008 to December 31, 2008. For drinking water collected from 2005 to 2008, the average PFOA concentration in drinking water was 0.04–2.47 µg/L.
North Wales, Pennsylvania	Pennsylvania Department of Health 2019	PFOS in Serum	Note: This site was not included in parameter estimates due to uncertainty regarding water concentrations. Serum levels for North Wales are reported in the PEATT Pilot Project, and the methodology and study description are described in further detail for Horsham, PA above. Of note, there were 18 consumers of public water being serviced by both Warrington Township Water and Sewer, and North Wales Water Authority. The mean serum PFOS level of public water drinkers serviced by the Warrington Township and the North Wales Water Authority (n=18) was 4.53 µg/L (95% CI: 3.51–5.85) in May–September 2018.

Location	Source	Data extracted	Study description
North Wales, Pennsylvania	NWWA 2018	PFOS in Water	Note: This site was dropped from the analysis due to uncertainty regarding water concentrations. In 2019, the North Wales Water Authority completed its acquisition of the Warrington Township Water and Sewer. As stated below, Warrington reported a PFOS concentration of 0.0015 µg/L (range: 0 – 0.0023 µg/L) in its 2018 water quality report. The North Wales Water Authority's 2018 Water Quality Report reports a PFOS concentration of 0.0015 µg/L (0–0.0023 µg/L) in 2018. The same water concentration was reported in both water supplies; therefore, it may be reasonable to assume that households serviced by both water systems were likely exposed to .0015 µg/L of PFOS in their drinking water. However, we excluded this study due to uncertainty regarding timing and distribution of water concentrations among the 18 consumers with measured serum concentrations.
Paulsboro, New Jersey	Graber et al. 2019	PFOS in Serum	Details on participant recruitment and blood testing for this study are described above for PFOA. For the 164 participants who were more than 12 years of age, the geometric mean PFOS serum concentrations were 5.37 µg/L (95% CI: 4.75, 6.06) in 2016.
Paulsboro, New Jersey	Post et al. 2013	PFOS in Water	This study details the results of raw water sampling of 29 public water systems in New Jersey, conducted by the New Jersey Department of Environmental Protection from August 2009 to February 2010. Community groundwater from the Paulsboro Water Department was sampled for 10 PFASs. Samples were taken from a designated raw water sample tap in Paulsboro. The testing found a PFOS water concentration of 0.001 µg/L in Paulsboro from August 2009 to February 2010. Additionally, testing was done at the same site in the Paulsboro community in 2013. Raw water sampled on September 17, 2013 had a PFOS concentration of 0.006 µg/L. Treated water samples from the same source were collected on the same day and had a PFOS water concentration of 0.0074 µg/L.

Location	Source	Data extracted	Study description
Ronneby, Sweden	Li et al. 2018	PFOS in Water and serum	Note: This study was used to inform the half-life priors. However, summary statistics for serum concentrations reported in the manuscript combined samples taken from pre- and post- intervention time periods, so these were not included in training or testing datasets for parameter estimation. The study methodology and details are described above under PFOA. PFOS water concentrations and human serum levels were also reported. Outgoing drinking water from Brantafors, which serviced one-third of Ronneby households, had a PFOS concentration of 8 µg/L on December 16, 2013. On the same day outgoing drinking water from Karragarden, servicing all other households had a PFOS concentration of 0.027 µg/L. In June 2014, the panel study group had a mean serum PFOS concentration among all participants of 387 µg/L ± 259 (min: 24.1, median: 345, max: 1500). Mean half-life for PFOS among all participants was 3.4 years, among men aged 15-50 years was 4.6 years, and among women aged 15-50 years was 3.1 years. There was a strong age-related effect with 1.4% increase in PFOS serum concentrations per year of age. At testing in September 2016, the mean PFOS serum decreased by 35% compared to June 2014. Water and serum PFOS concentrations for this panel study in Brantafors are summarized below: Serum and water PFOS concentration in Brantafors, Ronneby Panel Study participants (n=106) The “Main Ronneby” group of 3418 residents that was recruited from 2014 to 2016 from the entire Ronneby municipality had baseline mean serum PFOS concentrations of 245 µg/L ± 234 (min: 0.58, median: 176, max: 1870). A reference group of 242 residents from an unexposed municipality (Karlshamn) was examined in 2016 and had mean serum PFOS concentrations of 5.68 µg/L ± 6.19 (min: <0.5, median: 4.21, max: 55.3). The Main Ronneby group had median serum PFOS concentrations 42 times higher than the reference group.
Uppsala County, Sweden	Gyllenhammar et al. 2019	PFOS in Water and serum	Note: This study was excluded from training and testing datasets for parameter estimation because it reports cumulative time of drinking water exposures based on location and assumptions regarding contributions of source water from contaminated wells. Study details and methods for Uppsala, Sweden, are described further under PFOA.
Warminster, Pennsylvania	Pennsylvania Department of Health 2019	PFOS in Serum	Serum levels for Warminster, PA are reported in the PEATT Pilot Project, and the methodology and study description are provided in further detail for Horsham, PA above. Among public drinking water consumers serviced by the Warminster Municipal Authority (n=83), the mean serum PFOS concentration was 10.06 µg/L (95% CI: 7.89-12.83) between May to September 2018.
Warminster, Pennsylvania	EPA 2017	PFOS in Water	PFOS water concentration data was reported for the zip code 18974 – Warminster Municipal Authority. The earliest collection was made on November 19, 2013, and the latest collection was on August 19, 2014. Over this duration a total of 38 samples were taken and 10 samples detected PFOS. The mean and median concentration of PFOS was 0.28 and 0.15 µg/L respectively. The geometric mean concentration was detected as 0.15 µg/L with a GSD of 0.303.

Location	Source	Data extracted	Study description
Warrington, Pennsylvania	Pennsylvania Department of Health 2019	PFOS in Serum	Serum levels for Warrington, PA are reported in the PEATT Pilot Project, and the methodology and study description are provided in further detail for Horsham, PA above. Among public drinking water consumers serviced by only the Warrington Township Water and Sewer (n=31), the mean serum PFOS concentration was 12.39 µg/L (95% CI: 9.08-16.91) between May to September 2018.
Warrington, Pennsylvania	Warrington Township, 2014; Warrington Township, 2018	PFOS in Water	The Warrington Township produces an annual water quality report of the public water system. During 2014, the western portion of Warrington was serviced by North Wales Water Authority, and the eastern portion was serviced by nine wells. The eastern and western systems are interconnected. In 2014, the water PFOS concentration was 0.67 µg/L (range: 0 – 0.67 µg/L) from January 1 to December 31, 2014. During 2018, all water supplied by the Warrington Township was purchased from the North Wales Water Authority and Forest Park Water Treatment Plant. From January 1, 2018 to December 31, 2018, testing was conducted on the “Western and Eastern systems” of the Warrington Township water system. In 2018, the water PFOS level detected was 0.0015 ppt (range: 0–0.0023 µg/L). The source of contamination noted is firefighting foam and other man-made sources.
West Morgan/East Lawrence – Decatur, AL	ATSDR 2013	PFOS in Serum and water	Study description and details were provided in further detail above under PFOA. Between November 2008 and 2011 – levels can be presumed to be <0.4 µg/L (PFOA) and <0.2 µg/L (PFOS). Geometric mean PFOS serum levels of all 153 participants (not stratified by drinking water source) was 39.8 µg/L (range: 5.4 to 472 µg/L) in blood collected from April 12-15, 2010.
West Morgan/East Lawrence – Decatur, AL	ATSDR 2016	PFOS in Serum	Study description and details were provided in further detail above under PFOA. The data only reflect serum concentration change in participants of both the 2010 and 2016 investigations (n = 45). Total PFOS levels detected in 2010 (n=153) were 39.8 µg/L (geometric mean) and in 2016 (n=45) 23.4 µg/L (geometric mean).
PFHxS			
Horsham, Pennsylvania	Pennsylvania Department of Health 2019	PFHxS in Serum	Study description and details are provided for Horsham PA above under PFOA. The mean serum PFHxS level of public water drinkers serviced by the Horsham Water and Sewer Authority (n=61) was 8.90 µg/L (95% CI: 6.11 – 12.96) in May–September 2018.
Horsham, Pennsylvania	EPA 2017	PFHxS in Water	PFHxS water concentration data was reported for the zip code 19044 – Horsham Water and Sewer Authority. The earliest collection was made on 06/24/2014, and the latest collection was on 03/19/2015. Over this duration a total of 34 samples were taken, and 7 samples were used to detect the levels of PFHxS. The mean and median concentrations of PFHxS were 0.17 and 0.05 µg/L, respectively. The geometric mean concentration was detected as 0.091 µg/L with a GSD of 3.24.

Location	Source	Data extracted	Study description
North Wales, Pennsylvania	Pennsylvania Department of Health 2019	PFHxS in Serum	Note: These data were excluded from the model training and testing due to great uncertainty in the drinking water concentrations for this site. Serum levels for North Wales are reported in the PEATT Pilot Project, and the methodology and study description are provided in further detail for Horsham, PA above. Of note, there were 18 consumers of public water being serviced by both Warrington Township Water and Sewer, and North Wales Water Authority. The mean serum PFHxS level of public water drinkers serviced by the Warrington Township and the North Wales Water Authority (n=18) was 2.42 µg/L (95% CI: 1.55 – 3.79) in May – September 2018.
Paulsboro, New Jersey	Graber et al. 2019	PFHxS in Serum	Details on participant recruitment and blood testing for this study is described above for PFOA. PFHxS was detected in 124 participants who were 12 years of age and older. For these participants, the geometric mean PFHxS serum concentration was 2.03 µg/L (95% CI: 1.84 – 2.25) in 2016.
Paulsboro, New Jersey	Post et al. 2013	PFHxS in Water	This study details the results of raw water sampling of 29 public water systems in New Jersey, conducted by the New Jersey Department of Environmental Protection from August 2009 to February 2010. Community groundwater from the Paulsboro Water Department was sampled for 10 PFASs. Samples were taken from a designated raw water sample tap in Paulsboro. The testing found a PFHxS water concentration of <0.005 µg/L in Paulsboro from August 2009 to February 2010. Additionally, testing was done at the same site in the Paulsboro community in 2013. Raw water sampled on September 17, 2013 had a PFHxS concentration of 0.0044 µg/L. Treated water samples from the same source were collected on the same day and had a PFHxS water concentration of 0.0047 µg/L.
Uppsala County, Sweden	Gyllenhammar et al. 2019	PFHxS in Water and serum	Note: These data were excluded from the model training and testing as serum data due to lack of drinking water concentrations (cumulative metrics were reported). Study details and methods for Uppsala, Sweden are described further under PFOA.
Warminster, Pennsylvania	Pennsylvania Department of Health 2019	PFHxS in Serum	Serum levels for Warminster, PA are reported in the PEATT Pilot Project, and the methodology and study description are provided in further detail for Horsham, PA above. Among public drinking water consumers serviced by the Warminster Municipal Authority (n=83), the mean serum PFHxS concentration was 7.19 µg/L (95% CI: 5.31-9.73) between May and September 2018.
Warminster, Pennsylvania	EPA 2017	PFHxS in Water	PFHxS water concentration data were reported for the zip code 18974 – Warminster Municipal Authority. The earliest collection was made on November 19, 2013, and the latest collection was on August 19, 2014. Over this duration a total of 38 samples were taken, and 9 samples were used to detect the levels of PFHxS. The mean and median concentration of PFHxS was 0.127 and 0.076 µg/L respectively. The geometric mean concentration was detected as 0.083 µg/L with a GSD of 2.57.

Location	Source	Data extracted	Study description
Warrington, Pennsylvania	Pennsylvania Department of Health 2019	PFHxS in Serum	Serum levels for Warrington, PA are reported in the PEATT Pilot Project, and the methodology and study description are provided in further detail for Horsham, PA above. Among public drinking water consumers serviced by only the Warrington Township Water and Sewer (n=31), the mean serum PFHxS concentration was 7.69 µg/L (95% CI: 5.41-10.92) between May and September 2018.
PFNA			
Horsham, Pennsylvania	Pennsylvania Department of Health 2019b	PFNA in Serum	Study description and details are provided for Horsham PA above under PFOA. The mean serum PFNA level of public water drinkers serviced by the Horsham Water and Sewer Authority (n=61) was 0.76 µg/L (95% CI: 0.65 – 0.89) in May – September 2018.
Horsham, Pennsylvania	EPA 2017	PFNA in Water	PFNA water concentration data was reported for the zip code 19044 – Horsham Water and Sewer Authority. All samples taken were below the minimum reporting level of 0.02 µg/L. The earliest collection was made on June 24, 2014, and the latest collection was on March 19, 2015. Over this duration a total of 34 samples were taken.
Paulsboro, New Jersey	Graber et al. 2019	PFNA in Serum and water	Details on participant recruitment and blood testing for this study is described above for PFOA. Paulsboro, NJ had the highest groundwater concentrations of PFNA in the state at 0.096 µg/L during 2011 and 2012. In finished water samples taken at the same time, PFNA concentrations were 0.072 µg/L. In 165 participants aged 12 years and older from Paulsboro, NJ, the mean serum PFNA concentration in 2016 was 0.0035 µg/L (95% CI: 0.003, 0.004).
Warminster, Pennsylvania	Pennsylvania Department of Health 2019	PFNA in Serum	Serum levels for Warminster, PA are reported in the PEATT Pilot Project, and the methodology and study description are provided in further detail for Horsham, PA above. Among public drinking water consumers serviced by the Warminster Municipal Authority (n=83), the mean serum PFNA concentration was 0.72 µg/L (95% CI: 0.60-0.85) between May and September 2018.
Warminster, Pennsylvania	EPA 2017	PFNA in Water	PFNA water concentration data were reported for zip code 18974 – Warminster Municipal Authority. All samples taken were below the minimum reporting levels of 0.02 µg/L. The earliest collection was made on November 19, 2013, and the latest collection was on August 19, 2014. Over this duration a total of 38 samples were taken.

Note: CI = Confidence interval, GSD = Geometric Standard Deviation, LOD = limit of detection, MRL = Minimum Reporting Level, n = sample size, ND = nondetect, PFAS = perfluoroalkyl substance, PFOA = perfluorooctanoic acid, PFOS = perfluorooctane sulfonic acid, PFNA = perfluorononanoic acid, PFHXS = perfluorohexane sulfonate

Table S2. Water concentration source, level of data the concentration applies to, treatment of concentration over time, dates, concentration range, and data extraction methods for each location and chemical.

PF AS	Location	Source	Level	Over time	Dates	Range (µg/L)	Description
PFO A	WMEL – Decatur, AL, USA	ATSDR, 2016	Population	Varied	4/12/2010 - 1/1/2016	0.0173 - 0.0816	Mean of daily finished water at multiple time points extracted from plot.
PFO A	Arnsberg, Germany	Holzer et al., 2009	Population	Varied	10/10/2006 - 11/13/2007	0.01 - 0.23	Finished water over time extracted from plot.
PFO A	Lake Elmo/ Cottage Grove, MN, USA	Johnson et al., 2017	Individual	Fixed	01/01/05 - 12/31/08	0.035 - 1.9	Mean private well water extracted from plot.
PFO A	Little Hocking, OH, USA - Emmett	Emmett et al. 2006	Population	Fixed	01/01/02 - 05/31/05	3.55	Mean of finished water.
PFO A	Paulsboro, NJ, USA	Post et al., 2013	Population	Fixed	01/01/11 - 12/31/12	0.026	Mean of finished water.
PFO A	Horsham, PA, USA	EPA, 2017	Population	Fixed	06/24/14 - 03/19/15	0.075	Mean of finished water.
PFO A	Warminster, PA, USA	EPA, 2017	Population	Fixed	06/24/14 - 03/19/15	0.091	Mean of finished water.
PFO A	Warrington, PA, USA	EPA, 2017	Population	Fixed	06/24/14 - 03/19/15	0.12	Mean of finished water.
PFO A	Lubeck, OH, USA - Bartell - Public Water	Bartell, 2010; This analysis	Population	Fixed	05/01/07 - 06/30/07	0 - 0.016	Modeled below MRL
PFO A	Lubeck, OH, USA - Bartell - Bottled Water	Bartell, 2010; This analysis	Population	Fixed	NA	0 - 0.016	Modeled below MRL
PFO A	Little Hocking, OH, USA - Bartell - Public Water	Bartell, 2010; This analysis	Population	Fixed	05/01/07 - 06/30/07	0 - 0.016	Modeled below MRL
PFO A	Little Hocking, OH, USA - Bartell - Bottled Water	Bartell, 2010; This analysis	Population	Fixed	NA	0 - 0.016	Modeled below MRL
PFO S	WMEL – Decatur, AL, USA	ATSDR, 2013; ATSDR, 2016	Population	Varied	4/12/2010 - 1/1/2016	0.028 - 0.13	Mean of finished water at multiple time points extracted from plot.
PFO S	Lake Elmo/ Cottage Grove, MN, USA	Johnson et al., 2017	Individual	Fixed	01/01/05 - 12/31/08	0.035 - 1.8	Private well water concentrations as extracted from plot.
PFO S	Paulsboro, NJ, USA	Post et al. 2013	Population	Fixed	08/01/09 - 02/28/10	0.01	Mean of finished water
PFO S	Horsham, PA, USA	EPA, 2017; Pennsylv	Population	Fixed	06/24/14 - 03/19/15	0.2996	Mean of finished water

PF AS	Location	Source	Level	Over time	Dates	Range (µg/L)	Description
		ania Department of Health 2019					
PFO S	Warminster, PA, USA	EPA, 2017	Population	Fixed	06/24/14 - 03/19/15	0.28	Mean of finished water
PFO S	Warrington, PA, USA	EPA, 2017	Population	Fixed	06/24/14 - 03/19/15	0.67	Mean of finished water
PFH xS	WMEL – Decatur, AL, USA	ATSDR, 2016; This analysis	Individual	Varied	4/12/2010 - 1/1/2016	0 – 0.03	Modeled below MRL
PFH xS	Horsham, PA, USA	EPA, 2017	Population	Fixed	06/24/14 - 03/19/15	0.17	Mean of finished water
PFH xS	Paulsboro, NJ, USA	Post et al., 2013	Population	Fixed	9/17/2013	0.0047	Single finished water sample
PFH xS	Warminster, PA, USA	EPA, 2017	Population	Fixed	06/24/14 - 03/19/15	0.127	Mean of finished water
PFH xS	Warrington, PA, USA	EWG, 2019	Population	Fixed	11/01/14 - 11/30/14	0.0349	Mean of finished water
PFN A	WMEL – Decatur, AL, USA	ATSDR, 2016; This analysis	Individual	Varied	4/12/2010 - 1/1/2016	0 – 0.02	Modeled below MRL
PFN A	Horsham, PA, USA	EPA, 2017; This analysis	Population	Fixed	4/12/2010 - 1/1/2016	0 – 0.02	Modeled below MRL
PFN A	Paulsboro, NJ, USA	Post et al., 2013	Population	Fixed	01/01/11 - 12/31/12	0.072	Mean of finished water
PFN A	Warminster, PA, USA	EPA, 2017; This analysis	Population	Fixed	06/24/14 - 03/19/15	0 – 0.02	Modeled below MRL

Notes: Wherever possible, individual sample concentrations of finished water were used for each timepoint within range of measured serum concentrations. Where a date had more than one finished water sample, the arithmetic mean was taken. Arithmetic means of finished water were used when that was the only reported value. In cases where values were below the MRL, or where a population drank bottled water, we estimated water concentration as part of model training. In these cases, the range in the table is the range of the uniform prior used in estimation. All drinking water values used in the modeling can be found in the repository with the code. Abbreviations: AL = Alabama, MN = Minnesota, MRL = Minimum Reporting Level, NA = not available, NJ = New Jersey, OH = Ohio, PA = Pennsylvania, PFAS = perfluoroalkyl substance, PFOA = perfluorooctanoic acid, PFOS = perfluorooctane sulfonic acid, PFNA = perfluorononanoic acid, PFHXS = perfluorohexane sulfonate, WMEL = West Morgan/East Lawrence.

Table S3. Year-specific prior distributions for PFAS background concentrations (C_{bgd} ($\mu\text{g/L}$)) used for each study/population based on NHANES.

City/study – sex	Serum dates	NHANES population	NHANES years	PFOA GM	PFOA GSD	PFOS GM	PFOS GSD	PFNA GM	PFNA GSD	PFHxS GM	PFHxS GSD
Decatur/ATSDR 2013 – M	2010-2016	M 12+	2015-2016	1.80	1.81	6.16	2.18	0.63	1.93	1.61	2.17
Decatur/ATSDR 2013 – F	2010-2016	F 12+	2015-2016	1.36	2.14	3.67	2.39	0.53	2.18	0.88	2.47
Arnsberg/Hölzer et al. 2008 – M	2006-2007	M 18-69	2006-2007	4.92	1.70	N/A	N/A	N/A	N/A	N/A	N/A
Arnsberg/Hölzer et al. 2008 – F	2006-2007	F 23-49	2006-2007	3.07	1.85	N/A	N/A	N/A	N/A	N/A	N/A
Minnesota/MNEPHT – Mixed	2008	All 20+	2007-2008	4.15	1.69	13.47	2.21	N/A	N/A	N/A	N/A
Lubeck/Bartell et al. 2010 – Mixed	2007-2008	All 18+	2006-2007	4.15	1.83	N/A	N/A	N/A	N/A	N/A	N/A
Little Hocking/Bartell et al. 2010 – Mixed	2007-2008	All 18+	2006-2007	4.15	1.83	N/A	N/A	N/A	N/A	N/A	N/A
Little Hocking/Emmett et al. 2010 – Mixed	2004-2005	All 2+	2003-2004	3.95	1.90	N/A	N/A	N/A	N/A	N/A	N/A
Paulsboro/Graber et al. 2019 – Mixed	2016	All 12+	2015-2016	1.56	2.01	4.72	2.38	0.58	2.07	1.18	2.45
Horsham/Pennsylvania Department of Health 2019 – Mixed	2018	All 12+	2015-2016	1.56	2.01	4.72	2.38	0.58	2.07	1.18	2.45
Warminster/Pennsylvania Department of Health 2019 – Mixed	2018	All 12+	2015-2016	1.56	2.01	4.72	2.38	0.58	2.07	1.18	2.45
Warrington/Pennsylvania Department of Health 2019 – Mixed	2018	All 12+	2015-2016	1.56	2.01	4.72	2.38	N/A	N/A	1.18	2.45

Note: M = Male, F = Female, NHANES = National Health and Nutrition Examination Survey, GM = Geometric mean, GSD: Geometric standard deviation, PFOA = perfluorooctanoic acid, PFOS = perfluorooctane sulfonic acid, PFNA = perfluorononanoic acid, PFHXS = perfluorohexane sulfonate, N/A: Not applicable (no serum data or data not used in analysis).

Table S4. Results of sensitivity analysis. Parameter and 95% confidence intervals.

Sensitivity Analysis (abbreviation)	PFAS	Half-life GM	Half-life GSD	Vd GM	Vd GSD
Primary analysis (Cbgd-80)	PFOA	3.14 [2.69-3.73]	1.57 [1.42-1.73]	0.43 [0.32-0.59]	1.12 [1.01-1.47]
Background RSC = 0% (Cbgd-100)	PFOA	3.09 [2.63-3.61]	1.57 [1.42-1.76]	0.44 [0.34-0.6]	1.12 [1.01-1.44]
Background RSC = 80% (Cbgd-20)	PFOA	3.58 [3.06-4.22]	1.6 [1.45-1.78]	0.4 [0.31-0.53]	1.12 [1.01-1.46]
Vd prior decreased 20% (Vd-d20)	PFOA	3.1 [2.64-3.63]	1.56 [1.42-1.75]	0.4 [0.3-0.54]	1.12 [1.01-1.43]
Vd prior increased 20% (Vd-i20)	PFOA	3.26 [2.74-3.87]	1.58 [1.44-1.79]	0.47 [0.33-0.64]	1.12 [1.01-1.42]
Use only individual data (indiv)	PFOA	3.17 [2.84-3.56]	1.56 [1.45-1.7]	0.55 [0.42-0.7]	1.12 [1.01-1.41]
Use only population data (pop)	PFOA	2.54 [1.74-4.25]	1.34 [1.24-1.49]	0.22 [0.13-0.34]	1.13 [1-1.51]
Use all data for training (all-train)	PFOA	3.17 [2.85-3.57]	1.55 [1.43-1.68]	0.52 [0.41-0.69]	1.13 [1-1.42]
Swap training and testing data (swap)	PFOA	2.9 [2.52-3.41]	1.43 [1.32-1.58]	0.44 [0.32-0.6]	1.15 [1.01-1.53]
Primary analysis (Cbgd-80)	PFOS	3.42 [2.62-4.5]	1.57 [1.43-1.75]	0.32 [0.22-0.47]	1.11 [1-1.41]
Background RSC = 0% (Cbgd-100)	PFOS	3.19 [2.4-4.35]	1.56 [1.43-1.74]	0.31 [0.21-0.48]	1.12 [1.01-1.43]
Background RSC = 80% (Cbgd-20)	PFOS	3.94 [3.05-5.05]	1.59 [1.44-1.75]	0.26 [0.17-0.4]	1.11 [1-1.41]
Vd prior decreased 20% (Vd-d20)	PFOS	3.1 [2.27-4.01]	1.54 [1.42-1.75]	0.26 [0.18-0.39]	1.11 [1.01-1.46]
Vd prior increased 20% (Vd-i20)	PFOS	3.5 [2.64-4.68]	1.57 [1.44-1.78]	0.35 [0.23-0.54]	1.11 [1-1.43]
Use only individual data (indiv)	PFOS	3.47 [2.71-4.28]	1.52 [1.43-1.64]	0.37 [0.26-0.52]	1.11 [1-1.38]
Use only population data (pop)	PFOS	2.87 [1.65-5.85]	1.34 [1.25-1.48]	0.26 [0.16-0.45]	1.12 [1.01-1.46]
Use all data for training (all-train)	PFOS	3.46 [2.71-4.25]	1.53 [1.41-1.66]	0.4 [0.28-0.64]	1.12 [1.01-1.41]
Swap training and testing data (swap)	PFOS	3.32 [2.57-4.27]	1.4 [1.31-1.53]	0.36 [0.23-0.55]	1.11 [1.01-1.39]
Primary analysis (Cbgd-80)	PFNA	2.35 [1.65-3.16]	1.53 [1.4-1.7]	0.19 [0.11-0.3]	1.12 [1.01-1.51]
Background RSC = 0% (Cbgd-100)	PFNA	2.18 [1.47-2.98]	1.54 [1.42-1.71]	0.19 [0.12-0.3]	1.12 [1-1.45]
Background RSC = 80% (Cbgd-20)	PFNA	2.61 [1.76-3.58]	1.56 [1.44-1.73]	0.18 [0.11-0.29]	1.12 [1.01-1.45]
Vd prior decreased 20% (Vd-d20)	PFNA	2.19 [1.52-3.04]	1.53 [1.4-1.71]	0.16 [0.1-0.24]	1.12 [1-1.47]
Vd prior increased 20% (Vd-i20)	PFNA	2.34 [1.62-3.09]	1.55 [1.42-1.75]	0.21 [0.14-0.33]	1.13 [1-1.5]
Use only individual data (indiv)	PFNA	2.38 [1.7-3.25]	1.52 [1.42-1.65]	0.19 [0.11-0.3]	1.12 [1.01-1.47]
Use only population data (pop)	PFNA	2.77 [1.6-5.98]	1.34 [1.25-1.51]	0.19 [0.12-0.31]	1.12 [1.01-1.43]
Use all data for training (all-train)	PFNA	2.38 [1.7-3.25]	1.52 [1.42-1.65]	0.19 [0.11-0.3]	1.12 [1.01-1.47]
Swap training and testing data (swap)	PFNA	2.73 [1.82-3.47]	1.42 [1.33-1.54]	0.19 [0.12-0.31]	1.11 [1-1.45]
Primary analysis (Cbgd-80)	PFHxS	8.3 [5.38-13.54]	1.57 [1.42-1.77]	0.29 [0.17-0.46]	1.11 [1-1.45]
Background RSC = 0% (Cbgd-100)	PFHxS	8.38 [5.51-12.78]	1.58 [1.45-1.77]	0.28 [0.17-0.45]	1.12 [1.01-1.41]
Background RSC = 80% (Cbgd-20)	PFHxS	9.53 [6.43-14.59]	1.6 [1.46-1.79]	0.26 [0.17-0.39]	1.12 [1.01-1.4]
Vd prior decreased 20% (Vd-d20)	PFHxS	8.35 [5.16-12.69]	1.56 [1.43-1.77]	0.23 [0.14-0.37]	1.12 [1-1.39]
Vd prior increased 20% (Vd-i20)	PFHxS	8.74 [6.1-13.71]	1.58 [1.44-1.73]	0.33 [0.2-0.5]	1.11 [1.01-1.44]
Use only individual data (indiv)	PFHxS	10.11 [7.25-15.13]	1.6 [1.47-1.77]	0.26 [0.15-0.42]	1.13 [1.01-1.45]
Use only population data (pop)	PFHxS	5.9 [3.43-9.68]	1.34 [1.23-1.49]	0.24 [0.14-0.38]	1.13 [1.01-1.5]
Use all data for training (all-train)	PFHxS	9.9 [6.97-14.5]	1.63 [1.47-1.81]	0.3 [0.19-0.47]	1.1 [1-1.41]
Swap training and testing data (swap)	PFHxS	8.27 [5.15-12.29]	1.47 [1.35-1.61]	0.26 [0.17-0.42]	1.12 [1.01-1.42]

Note: Cbgd = Background Serum concentrations, GM= Geometric mean, GSD= Geometric standard deviation, PFOA = perfluorooctanoic acid, PFOS = perfluorooctane sulfonic acid, PFNA = perfluorononanoic acid, PFHXS = perfluorohexane sulfonate, RSC = relative source concentration, Vd = Volume of distribution.

Table S5. Figure 2 fit statistics comparing a one to one relationship between measured and predicted serum concentrations for training and testing data.

PFAS	Training	Test
PFOA	$R^2 = 0.991$, RMSE = 0.033, n = 311	$R^2 = 0.678$, RMSE = 0.165, n = 313
PFOS	$R^2 = 0.998$, RMSE = 0.014, n = 87	$R^2 = 0.803$, RMSE = 0.119, n = 89
PFNA	$R^2 = 0.987$, RMSE = 0.046, n = 38	$R^2 = 0.987$, RMSE = 0.038, n = 39
PFHxS	$R^2 = 0.563$, RMSE = 0.235, n = 38	$R^2 = 0.807$, RMSE = 0.102, n = 40

Note: n = number of points compared to prediction; includes individual measurements, repeated measurements, and summary statistics for populations. RMSE = Root mean square error, PFAS = perfluoroalkyl substance, PFOA = perfluorooctanoic acid, PFOS = perfluorooctane sulfonic acid, PFNA = perfluorononanoic acid, PFHXS = perfluorohexane sulfonate

Table S6. Figure 3 summary statistics. Each row represents a single measurement from an individual, at a particular timepoint. Note: IQR = Interquartile Range.

PFAS	Dataset identifier	Years from initial serum measurement	Minimum	Lower Interquartile Range	Median	Upper Interquartile Range	Maximum	Position of measured value
PFOA	Train 10	0	6.025	7.231	7.687	8.111	9.423	lower IQR
PFOA	Train 10	5.802	4.268	5.219	5.523	5.853	6.759	lower IQR
PFOA	Train 11	0	36.963	44.845	47.674	50.545	59.044	lower IQR
PFOA	Train 11	5.802	22.556	27.771	29.809	31.782	37.735	upper IQR
PFOA	Train 12	0	8.700	10.417	11.055	11.706	13.553	lower IQR
PFOA	Train 12	5.802	7.325	8.889	9.528	10.162	11.915	upper IQR
PFOA	Train 13	0	5.369	6.430	6.742	7.163	8.205	lower IQR
PFOA	Train 13	5.802	5.411	6.737	7.236	7.690	8.998	upper IQR
PFOA	Train 14	0	51.377	62.853	66.237	70.773	82.334	lower IQR
PFOA	Train 14	5.802	24.808	29.356	31.149	32.802	37.902	upper IQR
PFOA	Train 15	0	41.093	49.173	52.459	55.435	64.554	below IQR
PFOA	Train 15	5.802	29.413	36.168	38.604	41.294	48.685	above IQR
PFOA	Train 16	0	10.811	12.414	13.088	13.866	16.033	lower IQR
PFOA	Train 16	5.802	6.470	7.891	8.466	8.975	10.523	upper IQR
PFOA	Train 17	0	36.594	43.172	45.296	47.750	53.890	upper IQR
PFOA	Train 17	5.802	14.143	16.736	17.922	19.029	22.396	lower IQR
PFOA	Train 18	0	27.113	32.294	34.353	36.443	42.409	lower IQR
PFOA	Train 18	5.802	11.552	14.113	14.972	16.129	18.626	upper IQR
PFOA	Train 1	0	34.507	40.519	42.869	45.154	52.026	lower IQR
PFOA	Train 1	5.802	21.698	26.038	27.854	29.901	35.431	upper IQR
PFOA	Train 2	0	10.687	12.445	13.224	14.101	16.432	lower IQR
PFOA	Train 2	5.802	7.387	9.416	10.063	10.806	12.832	upper IQR
PFOA	Train 3	0	19.658	24.097	25.472	27.076	31.267	upper IQR
PFOA	Train 3	5.802	7.861	9.486	10.203	10.872	12.812	lower IQR
PFOA	Train 4	0	37.583	44.537	47.260	49.899	57.736	upper IQR
PFOA	Train 4	5.802	8.459	10.867	11.598	12.520	14.842	lower IQR

PFOA	Dataset identifier	Years from initial serum measurement	Minimum	Lower Interquartile Range	Median	Upper Interquartile Range	Maximum	Position of measured value
PFOA	Train 5	0	41.006	48.648	51.328	54.235	62.378	lower IQR
PFOA	Train 5	5.802	14.816	18.090	19.248	20.514	23.975	upper IQR
PFOA	Train 6	0	10.190	12.067	12.702	13.390	15.274	lower IQR
PFOA	Train 6	5.802	7.604	9.120	9.710	10.269	11.964	upper IQR
PFOA	Train 7	0	33.756	41.199	43.700	46.415	53.836	upper IQR
PFOA	Train 7	5.802	10.135	11.971	12.872	13.893	16.511	lower IQR
PFOA	Train 8	0	36.933	44.490	47.183	50.420	58.477	upper IQR
PFOA	Train 8	5.802	12.063	14.787	15.717	16.624	18.948	lower IQR
PFOA	Train 9	0	15.807	18.953	19.891	21.061	23.710	lower IQR
PFOA	Train 9	5.802	9.640	11.702	12.430	13.273	15.593	upper IQR
PFOA	Test 138	0	4.054	9.372	12.171	16.160	25.477	upper IQR
PFOA	Test 138	5.802	1.016	4.076	5.898	8.313	14.495	lower IQR
PFOA	Test 139	0	13.516	32.227	42.901	56.461	92.474	upper IQR
PFOA	Test 139	5.802	0.999	9.446	13.872	19.883	35.270	above IQR
PFOA	Test 140	0	3.664	11.559	15.155	19.804	32.133	upper IQR
PFOA	Test 140	5.802	0.702	4.696	6.766	9.443	16.476	upper IQR
PFOA	Test 141	0	5.633	13.968	18.142	24.122	38.838	upper IQR
PFOA	Test 141	5.802	0.971	4.936	7.777	10.656	18.591	above IQR
PFOA	Test 142	0	5.787	14.256	18.715	23.935	38.205	lower IQR
PFOA	Test 142	5.802	0.985	5.594	7.976	11.103	19.363	upper IQR
PFOA	Test 143	0	6.390	16.994	22.748	30.342	50.094	upper IQR
PFOA	Test 143	5.802	0.753	5.721	8.616	12.448	22.365	below IQR
PFOA	Test 144	0	4.630	14.477	19.247	25.366	41.549	upper IQR
PFOA	Test 144	5.802	1.222	5.300	7.950	10.828	19.118	above IQR
PFOA	Test 145	0	13.119	30.457	39.978	55.272	92.354	upper IQR
PFOA	Test 145	5.802	0.638	8.604	12.861	19.157	34.170	upper IQR
PFOA	Test 146	0	3.374	8.989	11.484	14.953	23.800	upper IQR
PFOA	Test 146	5.802	0.556	3.939	5.637	7.850	13.648	upper IQR

PFOA	Dataset identifier	Years from initial serum measurement	Minimum	Lower Interquartile Range	Median	Upper Interquartile Range	Maximum	Position of measured value
PFOA	Test 147	0	5.738	12.905	17.596	23.319	38.918	upper IQR
PFOA	Test 147	5.802	0.999	5.307	7.456	10.572	17.842	upper IQR
PFOA	Test 129	0	2.959	6.304	8.557	11.509	18.801	upper IQR
PFOA	Test 129	5.802	1.120	3.779	5.297	7.311	12.519	below IQR
PFOA	Test 130	0	9.348	22.368	28.856	38.894	62.272	upper IQR
PFOA	Test 130	5.802	1.767	7.507	11.095	16.021	27.743	upper IQR
PFOA	Test 131	0	17.145	38.612	49.927	66.984	109.312	upper IQR
PFOA	Test 131	5.802	1.967	10.584	16.532	24.632	44.915	upper IQR
PFOA	Test 132	0	5.126	13.850	18.415	24.257	39.358	upper IQR
PFOA	Test 132	5.802	0.877	5.441	8.127	11.501	20.404	upper IQR
PFOA	Test 133	0	8.954	21.135	28.047	37.505	61.956	upper IQR
PFOA	Test 133	5.802	1.054	7.242	10.850	15.007	26.364	above IQR
PFOA	Test 134	0	7.590	21.049	28.172	37.086	59.794	upper IQR
PFOA	Test 134	5.802	1.090	7.043	10.193	15.345	27.392	above IQR
PFOA	Test 135	0	5.008	13.646	17.583	24.080	39.339	upper IQR
PFOA	Test 135	5.802	1.585	5.290	7.291	10.756	18.937	upper IQR
PFOA	Test 136	0	6.986	15.825	21.557	29.009	48.413	upper IQR
PFOA	Test 136	5.802	1.322	5.943	8.972	12.496	21.777	upper IQR
PFOA	Test 137	0	8.703	20.886	26.805	35.454	57.291	upper IQR
PFOA	Test 137	5.802	1.692	7.071	10.316	14.196	24.810	upper IQR
PFOS	Train 10	0	61.771	75.927	80.561	86.025	99.643	upper IQR
PFOS	Train 10	5.802	22.306	29.801	32.283	35.014	42.246	lower IQR
PFOS	Train 11	0	57.355	68.203	72.419	77.519	91.095	lower IQR
PFOS	Train 11	5.802	35.318	44.494	47.830	51.187	60.760	upper IQR
PFOS	Train 12	0	10.082	12.819	13.808	14.738	17.314	lower IQR
PFOS	Train 12	5.802	9.220	11.727	12.591	13.591	16.298	upper IQR
PFOS	Train 13	0	32.239	39.777	42.730	45.536	53.765	lower IQR
PFOS	Train 13	5.802	19.802	25.260	27.125	29.330	35.433	upper IQR

PFAS	Dataset identifier	Years from initial serum measurement	Minimum	Lower Interquartile Range	Median	Upper Interquartile Range	Maximum	Position of measured value
PFOS	Train 14	0	74.349	91.268	97.257	103.799	122.120	upper IQR
PFOS	Train 14	5.802	25.596	32.610	35.339	38.326	46.091	lower IQR
PFOS	Train 15	0	44.028	54.702	58.968	62.030	72.851	lower IQR
PFOS	Train 15	5.802	31.768	40.024	43.033	46.264	54.931	upper IQR
PFOS	Train 16	0	23.669	30.106	32.133	34.430	40.800	upper IQR
PFOS	Train 16	5.802	10.455	12.818	13.771	14.861	17.828	lower IQR
PFOS	Train 17	0	46.403	57.033	60.463	64.881	76.381	upper IQR
PFOS	Train 17	5.802	19.633	25.512	27.429	29.971	36.447	upper IQR
PFOS	Train 18	0	32.190	41.589	44.783	48.039	57.221	lower IQR
PFOS	Train 18	5.802	23.956	30.803	32.930	35.653	42.724	upper IQR
PFOS	Train 1	0	65.711	80.788	86.367	92.403	109.820	lower IQR
PFOS	Train 1	5.802	46.001	60.233	64.834	69.801	83.660	above IQR
PFOS	Train 2	0	23.270	29.916	32.258	34.366	40.617	upper IQR
PFOS	Train 2	5.802	10.768	13.578	14.610	15.646	18.371	lower IQR
PFOS	Train 3	0	176.071	220.086	234.920	250.264	295.347	upper IQR
PFOS	Train 3	5.802	55.286	69.959	75.607	81.598	98.398	lower IQR
PFOS	Train 4	0	45.585	56.348	59.875	63.957	74.800	upper IQR
PFOS	Train 4	5.802	9.923	12.379	13.260	14.400	17.313	lower IQR
PFOS	Train 5	0	134.827	169.267	180.191	193.519	226.341	upper IQR
PFOS	Train 5	5.802	33.283	42.152	45.429	49.101	59.360	lower IQR
PFOS	Train 6	0	19.223	23.867	25.794	27.131	31.811	lower IQR
PFOS	Train 6	5.802	13.230	17.372	18.743	20.181	24.181	upper IQR
PFOS	Train 7	0	86.027	104.559	111.923	119.138	139.335	upper IQR
PFOS	Train 7	5.802	17.551	22.807	24.803	26.906	32.370	lower IQR
PFOS	Train 8	0	58.708	72.090	76.989	82.293	96.987	upper IQR
PFOS	Train 8	5.802	19.473	24.917	26.690	29.155	34.679	lower IQR
PFOS	Train 9	0	41.004	50.975	54.144	57.806	68.025	upper IQR
PFOS	Train 9	5.802	19.777	24.939	26.752	28.634	33.819	lower IQR

PFAS	Dataset identifier	Years from initial serum measurement	Minimum	Lower Interquartile Range	Median	Upper Interquartile Range	Maximum	Position of measured value
PFOS	Test 28	0	7.381	19.851	27.120	35.909	59.131	lower IQR
PFOS	Test 28	5.802	2.680	10.509	15.232	21.457	37.826	lower IQR
PFOS	Test 29	0	38.290	89.670	120.785	165.192	277.202	lower IQR
PFOS	Test 29	5.802	2.692	29.510	43.783	64.928	117.104	upper IQR
PFOS	Test 30	0	18.596	44.850	62.435	83.223	140.333	lower IQR
PFOS	Test 30	5.802	0.787	17.495	26.718	39.592	72.049	lower IQR
PFOS	Test 31	0	10.492	29.926	42.487	56.275	95.484	lower IQR
PFOS	Test 31	5.802	2.385	13.598	19.590	28.985	51.878	below IQR
PFOS	Test 32	0	13.725	28.601	40.370	53.583	88.104	lower IQR
PFOS	Test 32	5.802	2.671	13.519	19.618	28.488	50.090	below IQR
PFOS	Test 33	0	5.970	13.183	18.387	24.427	39.573	lower IQR
PFOS	Test 33	5.802	2.055	8.540	12.462	17.272	30.130	upper IQR
PFOS	Test 34	0	5.885	13.908	19.517	26.295	43.940	lower IQR
PFOS	Test 34	5.802	2.597	9.639	13.570	18.634	31.801	upper IQR
PFOS	Test 35	0	5.416	15.750	21.686	29.946	49.546	lower IQR
PFOS	Test 35	5.802	2.228	9.852	14.094	19.523	33.917	lower IQR
PFOS	Test 36	0	16.861	39.503	55.526	74.777	122.980	lower IQR
PFOS	Test 36	5.802	2.775	16.213	25.236	36.038	65.608	upper IQR
PFOS	Test 37	0	4.822	11.677	16.065	22.162	37.775	lower IQR
PFOS	Test 37	5.802	1.045	8.455	12.031	17.399	29.564	below IQR
PFOS	Test 19	0	19.087	48.115	66.513	89.045	146.765	lower IQR
PFOS	Test 19	5.802	4.861	20.563	29.615	43.061	72.906	below IQR
PFOS	Test 20	0	29.794	66.142	91.875	121.822	203.875	lower IQR
PFOS	Test 20	5.802	4.415	24.366	36.902	53.926	96.416	lower IQR
PFOS	Test 21	0	24.743	56.087	76.172	102.888	170.992	lower IQR
PFOS	Test 21	5.802	5.376	22.865	33.248	48.039	84.893	upper IQR
PFOS	Test 22	0	17.159	50.924	70.049	94.569	150.041	lower IQR
PFOS	Test 22	5.802	3.139	21.532	29.584	41.993	72.570	upper IQR

PFAS	Dataset identifier	Years from initial serum measurement	Minimum	Lower Interquartile Range	Median	Upper Interquartile Range	Maximum	Position of measured value
PFOS	Test 23	0	22.320	54.305	75.730	100.640	168.699	lower IQR
PFOS	Test 23	5.802	4.134	22.322	32.641	45.292	78.939	lower IQR
PFOS	Test 24	0	25.026	55.851	78.450	110.174	189.209	lower IQR
PFOS	Test 24	5.802	3.705	22.385	32.557	47.223	83.890	upper IQR
PFOS	Test 25	0	7.338	17.551	24.409	32.691	54.965	lower IQR
PFOS	Test 25	5.802	2.638	11.464	17.008	23.392	40.919	lower IQR
PFOS	Test 26	0	17.250	44.369	61.474	81.055	133.156	lower IQR
PFOS	Test 26	5.802	3.751	20.159	27.678	38.623	66.248	lower IQR
PFOS	Test 27	0	37.022	99.929	137.980	189.158	319.597	lower IQR
PFOS	Test 27	5.802	4.307	32.803	48.777	74.002	132.361	upper IQR
PFNA	Train 10	0	1.229	1.516	1.617	1.721	2.011	lower IQR
PFNA	Train 10	5.802	0.714	0.920	0.990	1.066	1.283	upper IQR
PFNA	Train 11	0	1.513	1.879	2.014	2.128	2.491	lower IQR
PFNA	Train 11	5.802	0.788	1.013	1.088	1.169	1.402	upper IQR
PFNA	Train 12	0	0.744	0.928	0.981	1.052	1.236	upper IQR
PFNA	Train 12	5.802	0.291	0.380	0.412	0.443	0.528	lower IQR
PFNA	Train 13	0	1.217	1.574	1.687	1.817	2.165	upper IQR
PFNA	Train 13	5.802	0.301	0.384	0.412	0.442	0.530	lower IQR
PFNA	Train 14	0	3.010	3.651	3.894	4.120	4.802	upper IQR
PFNA	Train 14	5.802	0.843	1.101	1.208	1.302	1.597	lower IQR
PFNA	Train 15	0	1.752	2.281	2.437	2.637	3.118	lower IQR
PFNA	Train 15	5.802	2.399	3.107	3.349	3.623	4.354	upper IQR
PFNA	Train 16	0	0.948	1.127	1.199	1.279	1.504	upper IQR
PFNA	Train 16	5.802	0.435	0.559	0.606	0.653	0.792	lower IQR
PFNA	Train 17	0	2.133	2.698	2.876	3.079	3.621	upper IQR
PFNA	Train 17	5.802	0.652	0.841	0.909	0.983	1.185	lower IQR
PFNA	Train 18	0	0.931	1.116	1.197	1.274	1.471	upper IQR
PFNA	Train 18	5.802	0.282	0.378	0.409	0.445	0.544	lower IQR

PFAS	Dataset identifier	Years from initial serum measurement	Minimum	Lower Interquartile Range	Median	Upper Interquartile Range	Maximum	Position of measured value
PFNA	Train 1	0	1.481	1.794	1.926	2.045	2.417	lower IQR
PFNA	Train 1	5.802	0.773	1.004	1.083	1.164	1.394	upper IQR
PFNA	Train 2	0	0.840	1.040	1.109	1.188	1.406	lower IQR
PFNA	Train 2	5.802	0.603	0.809	0.884	0.948	1.142	upper IQR
PFNA	Train 3	0	1.531	1.948	2.092	2.247	2.592	upper IQR
PFNA	Train 3	5.802	0.530	0.667	0.714	0.768	0.917	lower IQR
PFNA	Train 4	0	1.432	1.763	1.879	2.020	2.386	upper IQR
PFNA	Train 4	5.802	0.363	0.472	0.518	0.557	0.684	lower IQR
PFNA	Train 5	0	2.646	3.235	3.434	3.666	4.275	upper IQR
PFNA	Train 5	5.802	0.645	0.841	0.905	0.980	1.164	lower IQR
PFNA	Train 6	0	0.939	1.123	1.198	1.278	1.500	upper IQR
PFNA	Train 6	5.802	0.424	0.564	0.607	0.658	0.798	lower IQR
PFNA	Train 7	0	1.973	2.492	2.664	2.859	3.401	upper IQR
PFNA	Train 7	5.802	0.562	0.756	0.814	0.885	1.068	lower IQR
PFNA	Train 8	0	1.301	1.581	1.695	1.811	2.154	upper IQR
PFNA	Train 8	5.802	0.365	0.478	0.515	0.554	0.663	lower IQR
PFNA	Train 9	0	0.840	1.028	1.096	1.169	1.362	upper IQR
PFNA	Train 9	5.802	0.446	0.566	0.615	0.660	0.798	lower IQR
PFNA	Test 28	0	0.726	1.305	1.499	1.721	2.328	upper IQR
PFNA	Test 28	5.802	0.127	0.552	0.756	1.046	1.693	above IQR
PFNA	Test 29	0	1.814	2.698	3.034	3.461	4.602	upper IQR
PFNA	Test 29	5.802	0.172	0.728	1.045	1.447	2.503	above IQR
PFNA	Test 30	0	1.069	1.674	1.903	2.170	2.822	lower IQR
PFNA	Test 30	5.802	0.167	0.578	0.827	1.171	2.032	lower IQR
PFNA	Test 31	0	0.607	0.884	0.997	1.130	1.496	upper IQR
PFNA	Test 31	5.802	0.067	0.443	0.607	0.814	1.362	above IQR
PFNA	Test 32	0	0.715	1.048	1.216	1.366	1.802	lower IQR
PFNA	Test 32	5.802	0.120	0.459	0.632	0.911	1.579	below IQR

PFAS	Dataset identifier	Years from initial serum measurement	Minimum	Lower Interquartile Range	Median	Upper Interquartile Range	Maximum	Position of measured value
PFNA	Test 33	0	1.241	2.176	2.516	2.809	3.758	lower IQR
PFNA	Test 33	5.802	0.105	0.660	0.926	1.233	2.082	lower IQR
PFNA	Test 34	0	0.690	1.035	1.184	1.341	1.793	upper IQR
PFNA	Test 34	5.802	0.105	0.473	0.651	0.901	1.507	below IQR
PFNA	Test 35	0	1.303	2.044	2.319	2.630	3.473	lower IQR
PFNA	Test 35	5.802	0.078	0.645	0.911	1.249	2.069	lower IQR
PFNA	Test 36	0	0.816	1.305	1.482	1.689	2.190	upper IQR
PFNA	Test 36	5.802	0.129	0.534	0.726	0.971	1.610	below IQR
PFNA	Test 37	0	1.105	1.657	1.869	2.127	2.808	upper IQR
PFNA	Test 37	5.802	0.160	0.594	0.824	1.101	1.820	upper IQR
PFNA	Test 19	0	0.849	1.308	1.494	1.701	2.260	upper IQR
PFNA	Test 19	5.802	0.119	0.549	0.757	1.014	1.712	below IQR
PFNA	Test 20	0	1.827	2.872	3.231	3.711	4.936	upper IQR
PFNA	Test 20	5.802	0.183	0.765	1.114	1.459	2.420	lower IQR
PFNA	Test 21	0	1.053	1.578	1.809	2.073	2.798	lower IQR
PFNA	Test 21	5.802	0.155	0.593	0.841	1.117	1.900	lower IQR
PFNA	Test 22	0	1.000	1.397	1.602	1.795	2.355	lower IQR
PFNA	Test 22	5.802	0.200	0.585	0.763	1.066	1.765	lower IQR
PFNA	Test 23	0	2.509	3.822	4.276	4.927	6.579	upper IQR
PFNA	Test 23	5.802	0.141	0.855	1.273	1.740	3.052	upper IQR
PFNA	Test 24	0	0.956	1.490	1.707	1.938	2.577	lower IQR
PFNA	Test 24	5.802	0.078	0.574	0.823	1.099	1.840	above IQR
PFNA	Test 25	0	0.711	1.131	1.292	1.460	1.952	upper IQR
PFNA	Test 25	5.802	0.195	0.517	0.706	1.009	1.746	below IQR
PFNA	Test 26	0	1.752	2.603	2.978	3.398	4.570	upper IQR
PFNA	Test 26	5.802	0.063	0.770	1.047	1.439	2.437	lower IQR
PFNA	Test 27	0	1.741	2.778	3.125	3.561	4.641	lower IQR
PFNA	Test 27	5.802	0.180	0.756	1.106	1.491	2.521	upper IQR

PFAS	Dataset identifier	Years from initial serum measurement	Minimum	Lower Interquartile Range	Median	Upper Interquartile Range	Maximum	Position of measured value
PFHxS	Train 10	0	1.719	2.118	2.289	2.426	2.867	upper IQR
PFHxS	Train 10	5.802	1.915	2.366	2.545	2.737	3.291	lower IQR
PFHxS	Train 11	0	10.149	12.637	13.509	14.399	16.733	lower IQR
PFHxS	Train 11	5.802	9.671	11.939	12.840	13.803	16.559	upper IQR
PFHxS	Train 12	0	1.785	2.248	2.431	2.591	3.047	lower IQR
PFHxS	Train 12	5.802	2.422	3.025	3.273	3.541	4.277	upper IQR
PFHxS	Train 13	0	2.136	2.584	2.733	2.916	3.373	lower IQR
PFHxS	Train 13	5.802	2.834	3.657	3.940	4.237	5.080	upper IQR
PFHxS	Train 14	0	14.290	17.899	19.221	20.405	23.815	lower IQR
PFHxS	Train 14	5.802	12.650	15.930	17.058	18.224	21.460	above IQR
PFHxS	Train 15	0	10.438	12.614	13.454	14.198	16.509	lower IQR
PFHxS	Train 15	5.802	9.486	11.712	12.410	13.295	15.578	upper IQR
PFHxS	Train 16	0	2.545	3.119	3.347	3.543	4.161	lower IQR
PFHxS	Train 16	5.802	2.907	3.731	4.013	4.346	5.268	upper IQR
PFHxS	Train 17	0	13.144	15.589	16.691	17.761	20.860	upper IQR
PFHxS	Train 17	5.802	9.078	11.478	12.384	13.236	15.851	lower IQR
PFHxS	Train 18	0	6.590	8.047	8.554	9.120	10.557	upper IQR
PFHxS	Train 18	5.802	4.892	6.242	6.702	7.166	8.502	lower IQR
PFHxS	Train 1	0	12.046	15.115	16.154	17.299	20.366	lower IQR
PFHxS	Train 1	5.802	11.310	14.365	15.384	16.483	19.499	above IQR
PFHxS	Train 2	0	3.752	4.666	4.994	5.340	6.312	lower IQR
PFHxS	Train 2	5.802	4.334	5.484	5.931	6.337	7.574	upper IQR
PFHxS	Train 3	0	8.420	10.466	11.191	11.913	14.077	upper IQR
PFHxS	Train 3	5.802	6.052	7.607	8.187	8.754	10.469	lower IQR
PFHxS	Train 4	0	8.987	11.393	12.239	13.191	15.671	upper IQR
PFHxS	Train 4	5.802	4.297	5.646	6.120	6.620	8.064	below IQR
PFHxS	Train 5	0	15.260	19.009	20.322	21.617	25.406	upper IQR
PFHxS	Train 5	5.802	9.386	12.082	13.053	13.982	16.797	lower IQR

PFAS	Dataset identifier	Years from initial serum measurement	Minimum	Lower Interquartile Range	Median	Upper Interquartile Range	Maximum	Position of measured value
PFHxS	Train 6	0	4.765	5.928	6.349	6.789	8.061	lower IQR
PFHxS	Train 6	5.802	4.906	6.163	6.606	7.125	8.472	upper IQR
PFHxS	Train 7	0	19.370	22.799	24.424	25.914	30.500	lower IQR
PFHxS	Train 7	5.802	13.169	16.855	18.190	19.469	23.296	lower IQR
PFHxS	Train 8	0	19.417	23.207	24.717	26.219	30.659	lower IQR
PFHxS	Train 8	5.802	14.667	18.309	19.759	20.910	24.713	upper IQR
PFHxS	Train 9	0	6.162	7.490	8.009	8.544	9.987	lower IQR
PFHxS	Train 9	5.802	6.052	7.634	8.183	8.753	10.413	upper IQR
PFHxS	Test 28	0	2.201	3.382	3.846	4.348	5.689	lower IQR
PFHxS	Test 28	5.802	1.333	3.355	4.300	5.449	8.542	lower IQR
PFHxS	Test 29	0	13.284	19.302	22.036	25.241	34.056	upper IQR
PFHxS	Test 29	5.802	4.912	13.079	15.624	18.801	27.277	above IQR
PFHxS	Test 30	0	3.544	5.698	6.457	7.387	9.856	upper IQR
PFHxS	Test 30	5.802	1.811	4.822	5.942	7.500	11.505	lower IQR
PFHxS	Test 31	0	4.370	6.701	7.509	8.562	11.346	lower IQR
PFHxS	Test 31	5.802	1.623	5.335	6.518	8.261	12.537	upper IQR
PFHxS	Test 32	0	4.254	6.613	7.519	8.614	11.614	lower IQR
PFHxS	Test 32	5.802	2.172	5.528	6.668	8.271	12.382	upper IQR
PFHxS	Test 33	0	4.339	6.614	7.451	8.668	11.718	upper IQR
PFHxS	Test 33	5.802	1.403	5.536	6.748	8.447	11.895	below IQR
PFHxS	Test 34	0	3.040	5.196	5.883	6.725	8.533	upper IQR
PFHxS	Test 34	5.802	1.987	4.499	5.535	6.964	10.566	upper IQR
PFHxS	Test 35	0	10.436	16.326	18.511	20.986	26.984	lower IQR
PFHxS	Test 35	5.802	4.477	10.664	13.644	16.161	24.066	lower IQR
PFHxS	Test 36	0	2.995	4.623	5.283	5.987	7.997	upper IQR
PFHxS	Test 36	5.802	1.733	4.325	5.381	6.874	10.559	below IQR
PFHxS	Test 37	0	3.349	4.863	5.516	6.228	8.127	upper IQR
PFHxS	Test 37	5.802	1.832	4.371	5.402	6.919	10.208	lower IQR

PFAS	Dataset identifier	Years from initial serum measurement	Minimum	Lower Interquartile Range	Median	Upper Interquartile Range	Maximum	Position of measured value
PFHxS	Test 19	0	3.568	5.675	6.406	7.331	9.809	lower IQR
PFHxS	Test 19	5.802	2.327	5.147	6.313	7.780	11.636	below IQR
PFHxS	Test 20	0	3.860	6.285	7.196	8.171	10.909	lower IQR
PFHxS	Test 20	5.802	2.199	5.542	6.881	8.419	12.723	upper IQR
PFHxS	Test 21	0	9.844	17.039	19.186	21.880	29.005	upper IQR
PFHxS	Test 21	5.802	4.949	11.939	14.337	16.893	24.270	lower IQR
PFHxS	Test 22	0	4.081	6.512	7.418	8.471	11.213	upper IQR
PFHxS	Test 22	5.802	2.895	5.933	6.990	8.478	12.236	above IQR
PFHxS	Test 23	0	5.115	7.499	8.723	9.991	13.563	lower IQR
PFHxS	Test 23	5.802	2.971	6.209	7.671	9.305	13.695	upper IQR
PFHxS	Test 24	0	7.297	10.787	12.379	14.293	18.990	upper IQR
PFHxS	Test 24	5.802	3.143	8.541	10.239	12.160	17.518	above IQR
PFHxS	Test 25	0	4.944	7.997	8.996	10.072	13.075	lower IQR
PFHxS	Test 25	5.802	2.625	6.680	7.978	9.966	14.811	below IQR
PFHxS	Test 26	0	4.623	6.422	7.503	8.606	11.536	lower IQR
PFHxS	Test 26	5.802	2.479	5.568	6.908	8.554	13.026	lower IQR
PFHxS	Test 27	0	8.963	12.784	14.538	16.336	21.549	lower IQR
PFHxS	Test 27	5.802	3.880	9.463	11.195	13.534	19.480	above IQR

PFAS	Dataset	Years from initial serum measurement	Minimum	Lower Interquartile Range	Median	Upper Interquartile Range	Maximum	Position of measured value
PFOA	Train 10	0	6.025	7.231	7.687	8.111	9.423	lower IQR
PFOA	Train 10	5.802	4.268	5.219	5.523	5.853	6.759	lower IQR
PFOA	Train 11	0	36.963	44.845	47.674	50.545	59.044	lower IQR
PFOA	Train 11	5.802	22.556	27.771	29.809	31.782	37.735	upper IQR
PFOA	Train 12	0	8.700	10.417	11.055	11.706	13.553	lower IQR
PFOA	Train 12	5.802	7.325	8.889	9.528	10.162	11.915	upper IQR
PFOA	Train 13	0	5.369	6.430	6.742	7.163	8.205	lower IQR
PFOA	Train 13	5.802	5.411	6.737	7.236	7.690	8.998	upper IQR
PFOA	Train 14	0	51.377	62.853	66.237	70.773	82.334	lower IQR
PFOA	Train 14	5.802	24.808	29.356	31.149	32.802	37.902	upper IQR
PFOA	Train 15	0	41.093	49.173	52.459	55.435	64.554	below IQR
PFOA	Train 15	5.802	29.413	36.168	38.604	41.294	48.685	above IQR
PFOA	Train 16	0	10.811	12.414	13.088	13.866	16.033	lower IQR
PFOA	Train 16	5.802	6.470	7.891	8.466	8.975	10.523	upper IQR
PFOA	Train 17	0	36.594	43.172	45.296	47.750	53.890	upper IQR
PFOA	Train 17	5.802	14.143	16.736	17.922	19.029	22.396	lower IQR
PFOA	Train 18	0	27.113	32.294	34.353	36.443	42.409	lower IQR
PFOA	Train 18	5.802	11.552	14.113	14.972	16.129	18.626	upper IQR
PFOA	Train 1	0	34.507	40.519	42.869	45.154	52.026	lower IQR
PFOA	Train 1	5.802	21.698	26.038	27.854	29.901	35.431	upper IQR
PFOA	Train 2	0	10.687	12.445	13.224	14.101	16.432	lower IQR
PFOA	Train 2	5.802	7.387	9.416	10.063	10.806	12.832	upper IQR
PFOA	Train 3	0	19.658	24.097	25.472	27.076	31.267	upper IQR
PFOA	Train 3	5.802	7.861	9.486	10.203	10.872	12.812	lower IQR
PFOA	Train 4	0	37.583	44.537	47.260	49.899	57.736	upper IQR
PFOA	Train 4	5.802	8.459	10.867	11.598	12.520	14.842	lower IQR
PFOA	Train 5	0	41.006	48.648	51.328	54.235	62.378	lower IQR
PFOA	Train 5	5.802	14.816	18.090	19.248	20.514	23.975	upper IQR
PFOA	Train 6	0	10.190	12.067	12.702	13.390	15.274	lower IQR
PFOA	Train 6	5.802	7.604	9.120	9.710	10.269	11.964	upper IQR
PFOA	Train 7	0	33.756	41.199	43.700	46.415	53.836	upper IQR
PFOA	Train 7	5.802	10.135	11.971	12.872	13.893	16.511	lower IQR

PFOA	Train 8	0	36.933	44.490	47.183	50.420	58.477	upper IQR
PFOA	Train 8	5.802	12.063	14.787	15.717	16.624	18.948	lower IQR
PFOA	Train 9	0	15.807	18.953	19.891	21.061	23.710	lower IQR
PFOA	Train 9	5.802	9.640	11.702	12.430	13.273	15.593	upper IQR
PFOA	Test 138	0	4.054	9.372	12.171	16.160	25.477	upper IQR
PFOA	Test 138	5.802	1.016	4.076	5.898	8.313	14.495	lower IQR
PFOA	Test 139	0	13.516	32.227	42.901	56.461	92.474	upper IQR
PFOA	Test 139	5.802	0.999	9.446	13.872	19.883	35.270	above IQR
PFOA	Test 140	0	3.664	11.559	15.155	19.804	32.133	upper IQR
PFOA	Test 140	5.802	0.702	4.696	6.766	9.443	16.476	upper IQR
PFOA	Test 141	0	5.633	13.968	18.142	24.122	38.838	upper IQR
PFOA	Test 141	5.802	0.971	4.936	7.777	10.656	18.591	above IQR
PFOA	Test 142	0	5.787	14.256	18.715	23.935	38.205	lower IQR
PFOA	Test 142	5.802	0.985	5.594	7.976	11.103	19.363	upper IQR
PFOA	Test 143	0	6.390	16.994	22.748	30.342	50.094	upper IQR
PFOA	Test 143	5.802	0.753	5.721	8.616	12.448	22.365	below IQR
PFOA	Test 144	0	4.630	14.477	19.247	25.366	41.549	upper IQR
PFOA	Test 144	5.802	1.222	5.300	7.950	10.828	19.118	above IQR
PFOA	Test 145	0	13.119	30.457	39.978	55.272	92.354	upper IQR
PFOA	Test 145	5.802	0.638	8.604	12.861	19.157	34.170	upper IQR
PFOA	Test 146	0	3.374	8.989	11.484	14.953	23.800	upper IQR
PFOA	Test 146	5.802	0.556	3.939	5.637	7.850	13.648	upper IQR
PFOA	Test 147	0	5.738	12.905	17.596	23.319	38.918	upper IQR
PFOA	Test 147	5.802	0.999	5.307	7.456	10.572	17.842	upper IQR
PFOA	Test 129	0	2.959	6.304	8.557	11.509	18.801	upper IQR
PFOA	Test 129	5.802	1.120	3.779	5.297	7.311	12.519	below IQR
PFOA	Test 130	0	9.348	22.368	28.856	38.894	62.272	upper IQR
PFOA	Test 130	5.802	1.767	7.507	11.095	16.021	27.743	upper IQR
PFOA	Test 131	0	17.145	38.612	49.927	66.984	109.312	upper IQR
PFOA	Test 131	5.802	1.967	10.584	16.532	24.632	44.915	upper IQR
PFOA	Test 132	0	5.126	13.850	18.415	24.257	39.358	upper IQR

PFOA	Test 132	5.802	0.877	5.441	8.127	11.501	20.404	upper IQR
PFOA	Test 133	0	8.954	21.135	28.047	37.505	61.956	upper IQR
PFOA	Test 133	5.802	1.054	7.242	10.850	15.007	26.364	above IQR
PFOA	Test 134	0	7.590	21.049	28.172	37.086	59.794	upper IQR
PFOA	Test 134	5.802	1.090	7.043	10.193	15.345	27.392	above IQR
PFOA	Test 135	0	5.008	13.646	17.583	24.080	39.339	upper IQR
PFOA	Test 135	5.802	1.585	5.290	7.291	10.756	18.937	upper IQR
PFOA	Test 136	0	6.986	15.825	21.557	29.009	48.413	upper IQR
PFOA	Test 136	5.802	1.322	5.943	8.972	12.496	21.777	upper IQR
PFOA	Test 137	0	8.703	20.886	26.805	35.454	57.291	upper IQR
PFOA	Test 137	5.802	1.692	7.071	10.316	14.196	24.810	upper IQR
PFOS	Train 10	0	61.771	75.927	80.561	86.025	99.643	upper IQR
PFOS	Train 10	5.802	22.306	29.801	32.283	35.014	42.246	lower IQR
PFOS	Train 11	0	57.355	68.203	72.419	77.519	91.095	lower IQR
PFOS	Train 11	5.802	35.318	44.494	47.830	51.187	60.760	upper IQR
PFOS	Train 12	0	10.082	12.819	13.808	14.738	17.314	lower IQR
PFOS	Train 12	5.802	9.220	11.727	12.591	13.591	16.298	upper IQR
PFOS	Train 13	0	32.239	39.777	42.730	45.536	53.765	lower IQR
PFOS	Train 13	5.802	19.802	25.260	27.125	29.330	35.433	upper IQR
PFOS	Train 14	0	74.349	91.268	97.257	103.799	122.120	upper IQR
PFOS	Train 14	5.802	25.596	32.610	35.339	38.326	46.091	lower IQR
PFOS	Train 15	0	44.028	54.702	58.968	62.030	72.851	lower IQR
PFOS	Train 15	5.802	31.768	40.024	43.033	46.264	54.931	upper IQR
PFOS	Train 16	0	23.669	30.106	32.133	34.430	40.800	upper IQR
PFOS	Train 16	5.802	10.455	12.818	13.771	14.861	17.828	lower IQR
PFOS	Train 17	0	46.403	57.033	60.463	64.881	76.381	upper IQR
PFOS	Train 17	5.802	19.633	25.512	27.429	29.971	36.447	upper IQR
PFOS	Train 18	0	32.190	41.589	44.783	48.039	57.221	lower IQR
PFOS	Train 18	5.802	23.956	30.803	32.930	35.653	42.724	upper IQR
PFOS	Train 1	0	65.711	80.788	86.367	92.403	109.820	lower IQR
PFOS	Train 1	5.802	46.001	60.233	64.834	69.801	83.660	above IQR

PFOS	Train 2	0	23.270	29.916	32.258	34.366	40.617	upper IQR
PFOS	Train 2	5.802	10.768	13.578	14.610	15.646	18.371	lower IQR
PFOS	Train 3	0	176.071	220.086	234.920	250.264	295.347	upper IQR
PFOS	Train 3	5.802	55.286	69.959	75.607	81.598	98.398	lower IQR
PFOS	Train 4	0	45.585	56.348	59.875	63.957	74.800	upper IQR
PFOS	Train 4	5.802	9.923	12.379	13.260	14.400	17.313	lower IQR
PFOS	Train 5	0	134.827	169.267	180.191	193.519	226.341	upper IQR
PFOS	Train 5	5.802	33.283	42.152	45.429	49.101	59.360	lower IQR
PFOS	Train 6	0	19.223	23.867	25.794	27.131	31.811	lower IQR
PFOS	Train 6	5.802	13.230	17.372	18.743	20.181	24.181	upper IQR
PFOS	Train 7	0	86.027	104.559	111.923	119.138	139.335	upper IQR
PFOS	Train 7	5.802	17.551	22.807	24.803	26.906	32.370	lower IQR
PFOS	Train 8	0	58.708	72.090	76.989	82.293	96.987	upper IQR
PFOS	Train 8	5.802	19.473	24.917	26.690	29.155	34.679	lower IQR
PFOS	Train 9	0	41.004	50.975	54.144	57.806	68.025	upper IQR
PFOS	Train 9	5.802	19.777	24.939	26.752	28.634	33.819	lower IQR
PFOS	Test 28	0	7.381	19.851	27.120	35.909	59.131	lower IQR
PFOS	Test 28	5.802	2.680	10.509	15.232	21.457	37.826	lower IQR
PFOS	Test 29	0	38.290	89.670	120.785	165.192	277.202	lower IQR
PFOS	Test 29	5.802	2.692	29.510	43.783	64.928	117.104	upper IQR
PFOS	Test 30	0	18.596	44.850	62.435	83.223	140.333	lower IQR
PFOS	Test 30	5.802	0.787	17.495	26.718	39.592	72.049	lower IQR
PFOS	Test 31	0	10.492	29.926	42.487	56.275	95.484	lower IQR
PFOS	Test 31	5.802	2.385	13.598	19.590	28.985	51.878	below IQR
PFOS	Test 32	0	13.725	28.601	40.370	53.583	88.104	lower IQR
PFOS	Test 32	5.802	2.671	13.519	19.618	28.488	50.090	below IQR
PFOS	Test 33	0	5.970	13.183	18.387	24.427	39.573	lower IQR
PFOS	Test 33	5.802	2.055	8.540	12.462	17.272	30.130	upper IQR
PFOS	Test 34	0	5.885	13.908	19.517	26.295	43.940	lower IQR
PFOS	Test 34	5.802	2.597	9.639	13.570	18.634	31.801	upper IQR
PFOS	Test 35	0	5.416	15.750	21.686	29.946	49.546	lower IQR

PFOS	Test 35	5.802	2.228	9.852	14.094	19.523	33.917	lower IQR
PFOS	Test 36	0	16.861	39.503	55.526	74.777	122.980	lower IQR
PFOS	Test 36	5.802	2.775	16.213	25.236	36.038	65.608	upper IQR
PFOS	Test 37	0	4.822	11.677	16.065	22.162	37.775	lower IQR
PFOS	Test 37	5.802	1.045	8.455	12.031	17.399	29.564	below IQR
PFOS	Test 19	0	19.087	48.115	66.513	89.045	146.765	lower IQR
PFOS	Test 19	5.802	4.861	20.563	29.615	43.061	72.906	below IQR
PFOS	Test 20	0	29.794	66.142	91.875	121.822	203.875	lower IQR
PFOS	Test 20	5.802	4.415	24.366	36.902	53.926	96.416	lower IQR
PFOS	Test 21	0	24.743	56.087	76.172	102.888	170.992	lower IQR
PFOS	Test 21	5.802	5.376	22.865	33.248	48.039	84.893	upper IQR
PFOS	Test 22	0	17.159	50.924	70.049	94.569	150.041	lower IQR
PFOS	Test 22	5.802	3.139	21.532	29.584	41.993	72.570	upper IQR
PFOS	Test 23	0	22.320	54.305	75.730	100.640	168.699	lower IQR
PFOS	Test 23	5.802	4.134	22.322	32.641	45.292	78.939	lower IQR
PFOS	Test 24	0	25.026	55.851	78.450	110.174	189.209	lower IQR
PFOS	Test 24	5.802	3.705	22.385	32.557	47.223	83.890	upper IQR
PFOS	Test 25	0	7.338	17.551	24.409	32.691	54.965	lower IQR
PFOS	Test 25	5.802	2.638	11.464	17.008	23.392	40.919	lower IQR
PFOS	Test 26	0	17.250	44.369	61.474	81.055	133.156	lower IQR
PFOS	Test 26	5.802	3.751	20.159	27.678	38.623	66.248	lower IQR
PFOS	Test 27	0	37.022	99.929	137.980	189.158	319.597	lower IQR
PFOS	Test 27	5.802	4.307	32.803	48.777	74.002	132.361	upper IQR
PFNA	Train 10	0	1.229	1.516	1.617	1.721	2.011	lower IQR
PFNA	Train 10	5.802	0.714	0.920	0.990	1.066	1.283	upper IQR
PFNA	Train 11	0	1.513	1.879	2.014	2.128	2.491	lower IQR
PFNA	Train 11	5.802	0.788	1.013	1.088	1.169	1.402	upper IQR
PFNA	Train 12	0	0.744	0.928	0.981	1.052	1.236	upper IQR
PFNA	Train 12	5.802	0.291	0.380	0.412	0.443	0.528	lower IQR
PFNA	Train 13	0	1.217	1.574	1.687	1.817	2.165	upper IQR
PFNA	Train 13	5.802	0.301	0.384	0.412	0.442	0.530	lower IQR

PFNA	Train 14	0	3.010	3.651	3.894	4.120	4.802	upper IQR
PFNA	Train 14	5.802	0.843	1.101	1.208	1.302	1.597	lower IQR
PFNA	Train 15	0	1.752	2.281	2.437	2.637	3.118	lower IQR
PFNA	Train 15	5.802	2.399	3.107	3.349	3.623	4.354	upper IQR
PFNA	Train 16	0	0.948	1.127	1.199	1.279	1.504	upper IQR
PFNA	Train 16	5.802	0.435	0.559	0.606	0.653	0.792	lower IQR
PFNA	Train 17	0	2.133	2.698	2.876	3.079	3.621	upper IQR
PFNA	Train 17	5.802	0.652	0.841	0.909	0.983	1.185	lower IQR
PFNA	Train 18	0	0.931	1.116	1.197	1.274	1.471	upper IQR
PFNA	Train 18	5.802	0.282	0.378	0.409	0.445	0.544	lower IQR
PFNA	Train 1	0	1.481	1.794	1.926	2.045	2.417	lower IQR
PFNA	Train 1	5.802	0.773	1.004	1.083	1.164	1.394	upper IQR
PFNA	Train 2	0	0.840	1.040	1.109	1.188	1.406	lower IQR
PFNA	Train 2	5.802	0.603	0.809	0.884	0.948	1.142	upper IQR
PFNA	Train 3	0	1.531	1.948	2.092	2.247	2.592	upper IQR
PFNA	Train 3	5.802	0.530	0.667	0.714	0.768	0.917	lower IQR
PFNA	Train 4	0	1.432	1.763	1.879	2.020	2.386	upper IQR
PFNA	Train 4	5.802	0.363	0.472	0.518	0.557	0.684	lower IQR
PFNA	Train 5	0	2.646	3.235	3.434	3.666	4.275	upper IQR
PFNA	Train 5	5.802	0.645	0.841	0.905	0.980	1.164	lower IQR
PFNA	Train 6	0	0.939	1.123	1.198	1.278	1.500	upper IQR
PFNA	Train 6	5.802	0.424	0.564	0.607	0.658	0.798	lower IQR
PFNA	Train 7	0	1.973	2.492	2.664	2.859	3.401	upper IQR
PFNA	Train 7	5.802	0.562	0.756	0.814	0.885	1.068	lower IQR
PFNA	Train 8	0	1.301	1.581	1.695	1.811	2.154	upper IQR
PFNA	Train 8	5.802	0.365	0.478	0.515	0.554	0.663	lower IQR
PFNA	Train 9	0	0.840	1.028	1.096	1.169	1.362	upper IQR
PFNA	Train 9	5.802	0.446	0.566	0.615	0.660	0.798	lower IQR
PFNA	Test 28	0	0.726	1.305	1.499	1.721	2.328	upper IQR
PFNA	Test 28	5.802	0.127	0.552	0.756	1.046	1.693	above IQR
PFNA	Test 29	0	1.814	2.698	3.034	3.461	4.602	upper IQR

PFNA	Test 29	5.802	0.172	0.728	1.045	1.447	2.503	above IQR
PFNA	Test 30	0	1.069	1.674	1.903	2.170	2.822	lower IQR
PFNA	Test 30	5.802	0.167	0.578	0.827	1.171	2.032	lower IQR
PFNA	Test 31	0	0.607	0.884	0.997	1.130	1.496	upper IQR
PFNA	Test 31	5.802	0.067	0.443	0.607	0.814	1.362	above IQR
PFNA	Test 32	0	0.715	1.048	1.216	1.366	1.802	lower IQR
PFNA	Test 32	5.802	0.120	0.459	0.632	0.911	1.579	below IQR
PFNA	Test 33	0	1.241	2.176	2.516	2.809	3.758	lower IQR
PFNA	Test 33	5.802	0.105	0.660	0.926	1.233	2.082	lower IQR
PFNA	Test 34	0	0.690	1.035	1.184	1.341	1.793	upper IQR
PFNA	Test 34	5.802	0.105	0.473	0.651	0.901	1.507	below IQR
PFNA	Test 35	0	1.303	2.044	2.319	2.630	3.473	lower IQR
PFNA	Test 35	5.802	0.078	0.645	0.911	1.249	2.069	lower IQR
PFNA	Test 36	0	0.816	1.305	1.482	1.689	2.190	upper IQR
PFNA	Test 36	5.802	0.129	0.534	0.726	0.971	1.610	below IQR
PFNA	Test 37	0	1.105	1.657	1.869	2.127	2.808	upper IQR
PFNA	Test 37	5.802	0.160	0.594	0.824	1.101	1.820	upper IQR
PFNA	Test 19	0	0.849	1.308	1.494	1.701	2.260	upper IQR
PFNA	Test 19	5.802	0.119	0.549	0.757	1.014	1.712	below IQR
PFNA	Test 20	0	1.827	2.872	3.231	3.711	4.936	upper IQR
PFNA	Test 20	5.802	0.183	0.765	1.114	1.459	2.420	lower IQR
PFNA	Test 21	0	1.053	1.578	1.809	2.073	2.798	lower IQR
PFNA	Test 21	5.802	0.155	0.593	0.841	1.117	1.900	lower IQR
PFNA	Test 22	0	1.000	1.397	1.602	1.795	2.355	lower IQR
PFNA	Test 22	5.802	0.200	0.585	0.763	1.066	1.765	lower IQR
PFNA	Test 23	0	2.509	3.822	4.276	4.927	6.579	upper IQR
PFNA	Test 23	5.802	0.141	0.855	1.273	1.740	3.052	upper IQR
PFNA	Test 24	0	0.956	1.490	1.707	1.938	2.577	lower IQR
PFNA	Test 24	5.802	0.078	0.574	0.823	1.099	1.840	above IQR
PFNA	Test 25	0	0.711	1.131	1.292	1.460	1.952	upper IQR
PFNA	Test 25	5.802	0.195	0.517	0.706	1.009	1.746	below IQR

PFNA	Test 26	0	1.752	2.603	2.978	3.398	4.570	upper IQR
PFNA	Test 26	5.802	0.063	0.770	1.047	1.439	2.437	lower IQR
PFNA	Test 27	0	1.741	2.778	3.125	3.561	4.641	lower IQR
PFNA	Test 27	5.802	0.180	0.756	1.106	1.491	2.521	upper IQR
PFHxS	Train 10	0	1.719	2.118	2.289	2.426	2.867	upper IQR
PFHxS	Train 10	5.802	1.915	2.366	2.545	2.737	3.291	lower IQR
PFHxS	Train 11	0	10.149	12.637	13.509	14.399	16.733	lower IQR
PFHxS	Train 11	5.802	9.671	11.939	12.840	13.803	16.559	upper IQR
PFHxS	Train 12	0	1.785	2.248	2.431	2.591	3.047	lower IQR
PFHxS	Train 12	5.802	2.422	3.025	3.273	3.541	4.277	upper IQR
PFHxS	Train 13	0	2.136	2.584	2.733	2.916	3.373	lower IQR
PFHxS	Train 13	5.802	2.834	3.657	3.940	4.237	5.080	upper IQR
PFHxS	Train 14	0	14.290	17.899	19.221	20.405	23.815	lower IQR
PFHxS	Train 14	5.802	12.650	15.930	17.058	18.224	21.460	above IQR
PFHxS	Train 15	0	10.438	12.614	13.454	14.198	16.509	lower IQR
PFHxS	Train 15	5.802	9.486	11.712	12.410	13.295	15.578	upper IQR
PFHxS	Train 16	0	2.545	3.119	3.347	3.543	4.161	lower IQR
PFHxS	Train 16	5.802	2.907	3.731	4.013	4.346	5.268	upper IQR
PFHxS	Train 17	0	13.144	15.589	16.691	17.761	20.860	upper IQR
PFHxS	Train 17	5.802	9.078	11.478	12.384	13.236	15.851	lower IQR
PFHxS	Train 18	0	6.590	8.047	8.554	9.120	10.557	upper IQR
PFHxS	Train 18	5.802	4.892	6.242	6.702	7.166	8.502	lower IQR
PFHxS	Train 1	0	12.046	15.115	16.154	17.299	20.366	lower IQR
PFHxS	Train 1	5.802	11.310	14.365	15.384	16.483	19.499	above IQR
PFHxS	Train 2	0	3.752	4.666	4.994	5.340	6.312	lower IQR
PFHxS	Train 2	5.802	4.334	5.484	5.931	6.337	7.574	upper IQR
PFHxS	Train 3	0	8.420	10.466	11.191	11.913	14.077	upper IQR
PFHxS	Train 3	5.802	6.052	7.607	8.187	8.754	10.469	lower IQR
PFHxS	Train 4	0	8.987	11.393	12.239	13.191	15.671	upper IQR
PFHxS	Train 4	5.802	4.297	5.646	6.120	6.620	8.064	below IQR
PFHxS	Train 5	0	15.260	19.009	20.322	21.617	25.406	upper IQR

PFHxS	Train 5	5.802	9.386	12.082	13.053	13.982	16.797	lower IQR
PFHxS	Train 6	0	4.765	5.928	6.349	6.789	8.061	lower IQR
PFHxS	Train 6	5.802	4.906	6.163	6.606	7.125	8.472	upper IQR
PFHxS	Train 7	0	19.370	22.799	24.424	25.914	30.500	lower IQR
PFHxS	Train 7	5.802	13.169	16.855	18.190	19.469	23.296	lower IQR
PFHxS	Train 8	0	19.417	23.207	24.717	26.219	30.659	lower IQR
PFHxS	Train 8	5.802	14.667	18.309	19.759	20.910	24.713	upper IQR
PFHxS	Train 9	0	6.162	7.490	8.009	8.544	9.987	lower IQR
PFHxS	Train 9	5.802	6.052	7.634	8.183	8.753	10.413	upper IQR
PFHxS	Test 28	0	2.201	3.382	3.846	4.348	5.689	lower IQR
PFHxS	Test 28	5.802	1.333	3.355	4.300	5.449	8.542	lower IQR
PFHxS	Test 29	0	13.284	19.302	22.036	25.241	34.056	upper IQR
PFHxS	Test 29	5.802	4.912	13.079	15.624	18.801	27.277	above IQR
PFHxS	Test 30	0	3.544	5.698	6.457	7.387	9.856	upper IQR
PFHxS	Test 30	5.802	1.811	4.822	5.942	7.500	11.505	lower IQR
PFHxS	Test 31	0	4.370	6.701	7.509	8.562	11.346	lower IQR
PFHxS	Test 31	5.802	1.623	5.335	6.518	8.261	12.537	upper IQR
PFHxS	Test 32	0	4.254	6.613	7.519	8.614	11.614	lower IQR
PFHxS	Test 32	5.802	2.172	5.528	6.668	8.271	12.382	upper IQR
PFHxS	Test 33	0	4.339	6.614	7.451	8.668	11.718	upper IQR
PFHxS	Test 33	5.802	1.403	5.536	6.748	8.447	11.895	below IQR
PFHxS	Test 34	0	3.040	5.196	5.883	6.725	8.533	upper IQR
PFHxS	Test 34	5.802	1.987	4.499	5.535	6.964	10.566	upper IQR
PFHxS	Test 35	0	10.436	16.326	18.511	20.986	26.984	lower IQR
PFHxS	Test 35	5.802	4.477	10.664	13.644	16.161	24.066	lower IQR
PFHxS	Test 36	0	2.995	4.623	5.283	5.987	7.997	upper IQR
PFHxS	Test 36	5.802	1.733	4.325	5.381	6.874	10.559	below IQR
PFHxS	Test 37	0	3.349	4.863	5.516	6.228	8.127	upper IQR
PFHxS	Test 37	5.802	1.832	4.371	5.402	6.919	10.208	lower IQR
PFHxS	Test 19	0	3.568	5.675	6.406	7.331	9.809	lower IQR
PFHxS	Test 19	5.802	2.327	5.147	6.313	7.780	11.636	below IQR

PFHxS	Test 20	0	3.860	6.285	7.196	8.171	10.909	lower IQR
PFHxS	Test 20	5.802	2.199	5.542	6.881	8.419	12.723	upper IQR
PFHxS	Test 21	0	9.844	17.039	19.186	21.880	29.005	upper IQR
PFHxS	Test 21	5.802	4.949	11.939	14.337	16.893	24.270	lower IQR
PFHxS	Test 22	0	4.081	6.512	7.418	8.471	11.213	upper IQR
PFHxS	Test 22	5.802	2.895	5.933	6.990	8.478	12.236	above IQR
PFHxS	Test 23	0	5.115	7.499	8.723	9.991	13.563	lower IQR
PFHxS	Test 23	5.802	2.971	6.209	7.671	9.305	13.695	upper IQR
PFHxS	Test 24	0	7.297	10.787	12.379	14.293	18.990	upper IQR
PFHxS	Test 24	5.802	3.143	8.541	10.239	12.160	17.518	above IQR
PFHxS	Test 25	0	4.944	7.997	8.996	10.072	13.075	lower IQR
PFHxS	Test 25	5.802	2.625	6.680	7.978	9.966	14.811	below IQR
PFHxS	Test 26	0	4.623	6.422	7.503	8.606	11.536	lower IQR
PFHxS	Test 26	5.802	2.479	5.568	6.908	8.554	13.026	lower IQR
PFHxS	Test 27	0	8.963	12.784	14.538	16.336	21.549	lower IQR
PFHxS	Test 27	5.802	3.880	9.463	11.195	13.534	19.480	above IQR

Figure S1. Perfluorooctanoic acid (PFOA) serum concentrations as data (points) and predictions (boxplots) for studies other than Decatur, AL. These include Arnsberg (German), Minnesota (USA Lake Elmo/ Cottage Grove), Lubeck (USA OH), Little Hocking (USA OH; separated into two datasets Bartell et al. 2010 and Emmet et al. 2006), Paulsboro (USA NJ), Horsham (USA PA), Warminster (USA PA), and Warrington (USA PA). Where present, variable shading indicates time, in years, since the first serum sample. Samples from the same individuals are paired such that the two left-most bars indicate samples from one individual taken at two time (T in years) points (the initial time point is T = 0). This figure shows that the training data points nearly always fall within the interquartile range of the predicted serum concentrations. For testing data, most points fall within the interquartile range, and all points fall within the upper and lower bounds of estimates. More detailed tabular data can be found in the project repository <https://github.com/wachiuphd/2022-Bayes-PFAS-PK>

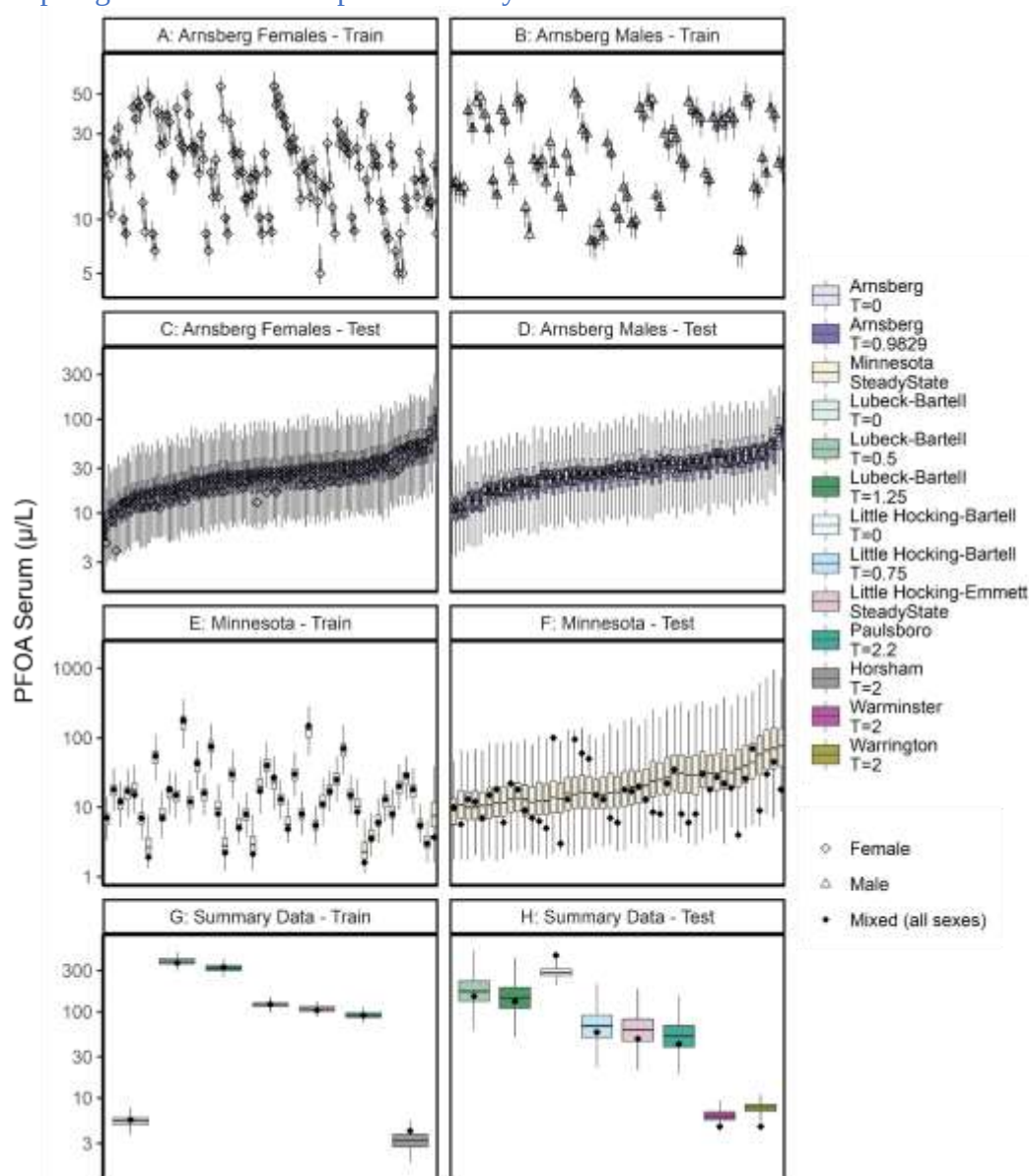


Figure S2. Perfluorooctanoic acid (PFOS) serum concentrations as data (points) and predictions (boxplots) for studies other than Decatur, AL. These include Minnesota (USA Lake Elmo/ Cottage Grove), Horsham (USA PA), Paulsboro (USA NJ), Warminster (USA PA), and Warrington (USA PA). Shading indicates that samples were all taken at a single time (T in years) point. This figure shows that the training data points generally fall within the interquartile range of the predicted serum concentrations. For testing data, most serum measurements (points) fall within the interquartile range, and few are outside of the upper and lower bounds of estimates. More detailed tabular data can be found in the project repository <https://github.com/wachiuphd/2022-Bayes-PFAS-PK>

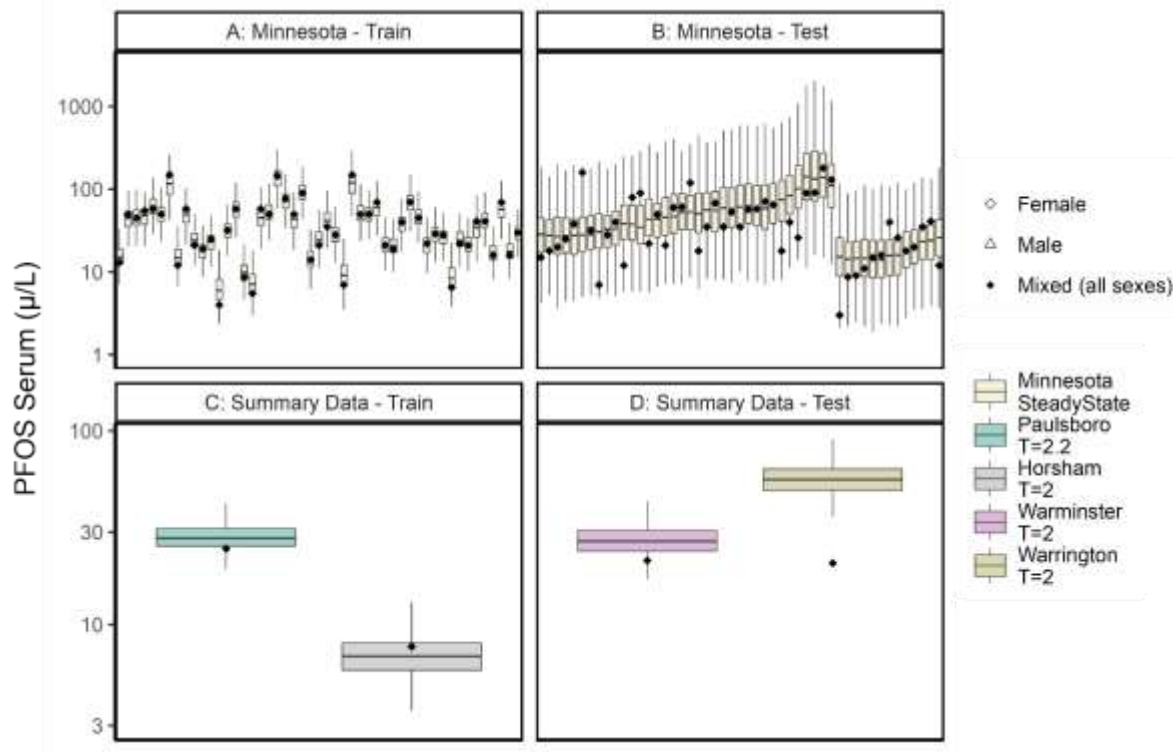


Figure S3. Perfluorononanoic acid (PFNA) data (points) and predictions (boxplots) for studies other than Decatur, AL. PFNA summary data included Paulsboro (USA NJ), Horsham (USA PA), and Warminster (USA PA). Shading indicates that samples were all taken at a single time (T in years) point. This figure shows that the all measured serum concentrations (points) fall within the interquartile range of the predicted serum for both training and testing data. More detailed tabular data can be found in the project repository <https://github.com/wachiuphd/2022-Bayes-PFAS-PK>

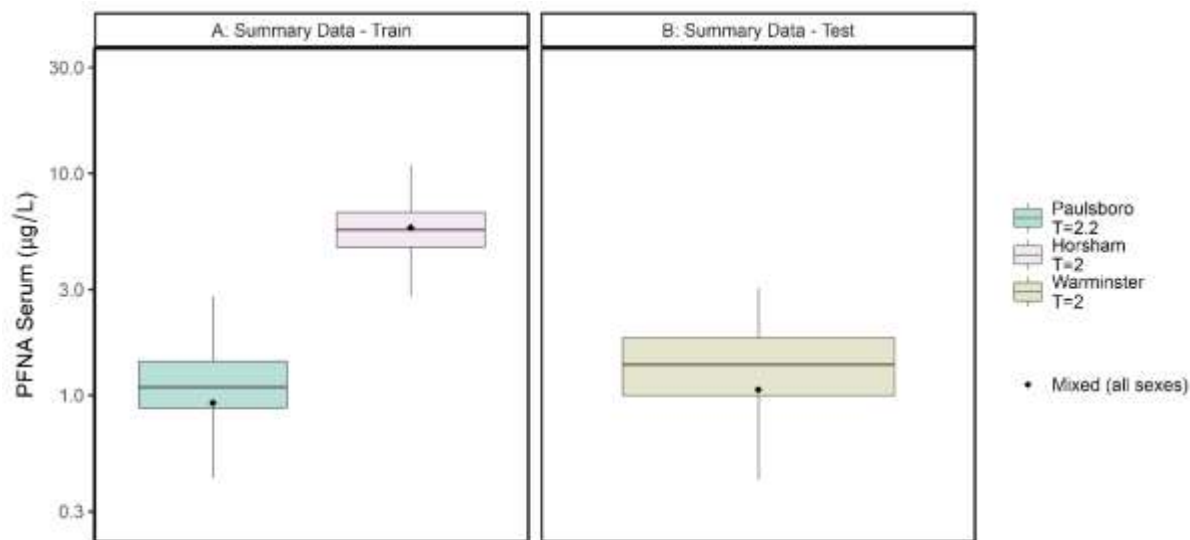


Figure S4. Perfluorohexane sulfonate (PFHxS) data (filled circles) and predictions (boxplots) for studies other than Decatur, AL. PFHxS summary data studies included Paulsboro (USA NJ), Horsham (USA PA), Warminster (USA PA), and Warrington (USA PA). Shading indicates that samples were all taken at a single time (T in years) point. This figure shows that the measured serum concentrations (points) fall within the range of the predicted serum concentrations for both training and testing data. More detailed tabular data can be found in the project repository <https://github.com/wachiuphd/2022-Bayes-PFAS-PK>

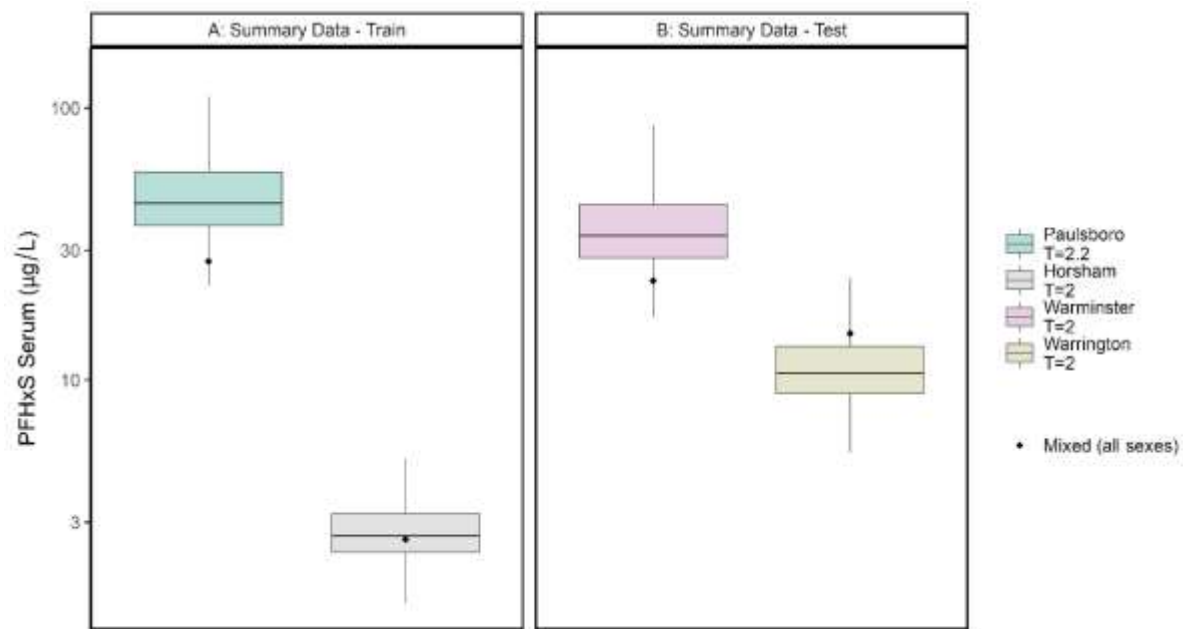


Figure S5. Comparison of V_d (left) and k (right) estimates across individuals from different datasets using the primary model. Specifically, for each individual a random estimate was drawn from the MCMC samples for their z-score for the corresponding parameter, and the resulting histogram compared to the expected standard normal distribution. Perfluorononanoic acid (PFNA) and perfluorohexane sulfonate (PFHxS) included only one individual-level dataset, so only comparisons for perfluorooctanoic acid (PFOA: top) and perfluorooctane sulfonic acid (PFOS: bottom) are included. There is no systematic effect of dataset origin (country or city) on V_d or k , and individual samples from each city were consistent with the expected standard normal distribution by one-sample Kolmogorov-Smirnov test (two-sided $p > 0.05$). More detailed information can be found in the project repository <https://github.com/wachiuphd/2022-Bayes-PFAS-PK>.

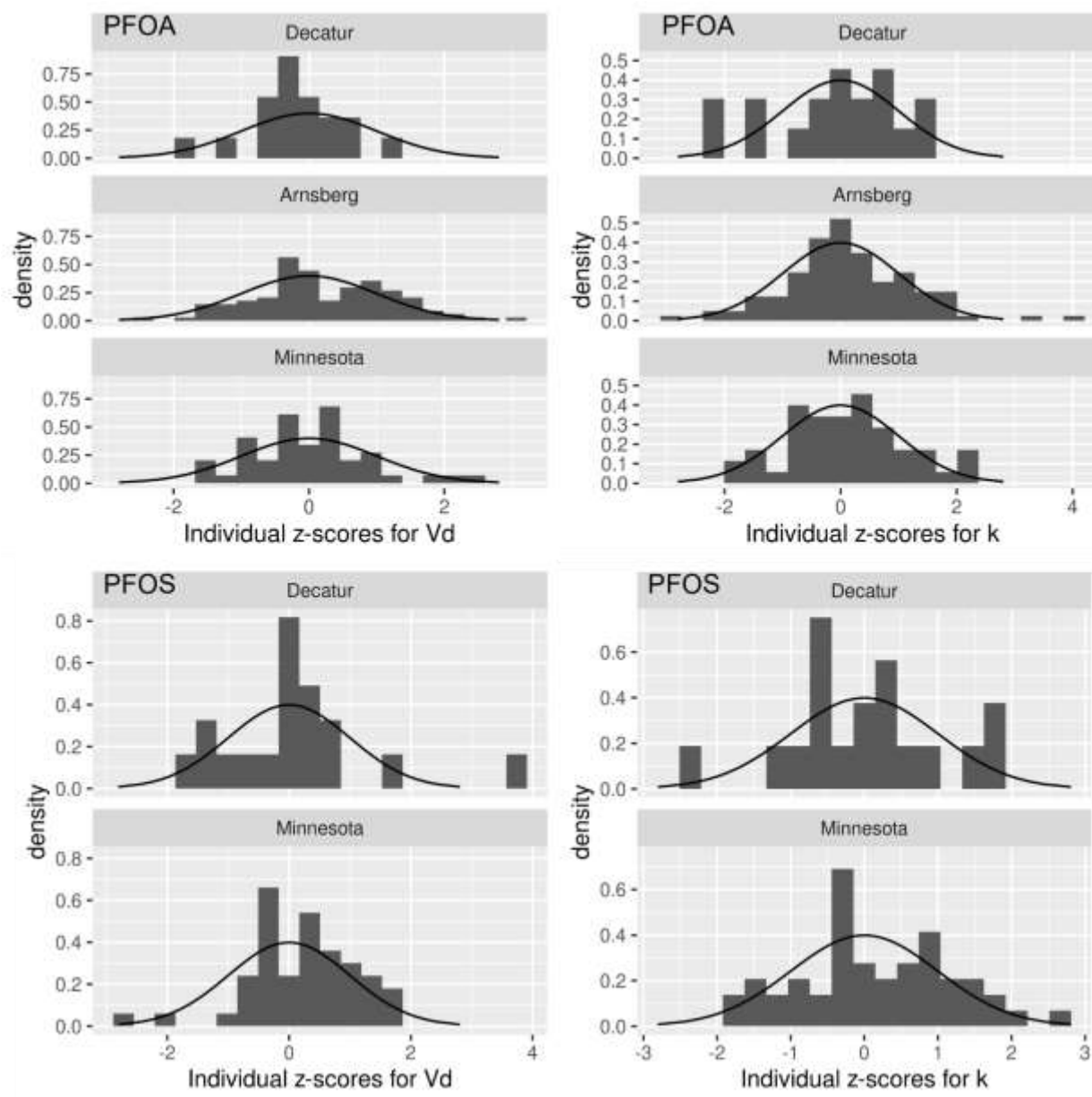


Figure S6. Results of sensitivity analysis showing posterior estimates of geometric mean of half-life (dotted line) for each of 9 alternative model definitions (described in methods and in Table S3). all-train = Use all data for training, Cbgd-80 = Primary analysis with background relative source contribution = 20%, Cbgd-100 = Background relative source contribution = 0%, Cbgd-20 = Background relative source contribution = 80%, Vd-d20 = Volume of distribution prior decreased 20%, Vd-i20 = Volume of distribution prior increased 20%, indiv = Use only individual data, pop = Use only population data, swap = Swap training and testing data, PFOA = perfluorooctanoic acid, PFOS = perfluorooctane sulfonic acid, PFNA = perfluorononanoic acid, PFHXS = perfluorohexane sulfonate. . More detailed information can be found in the project repository <https://github.com/wachiuphd/2022-Bayes-PFAS-PK>.

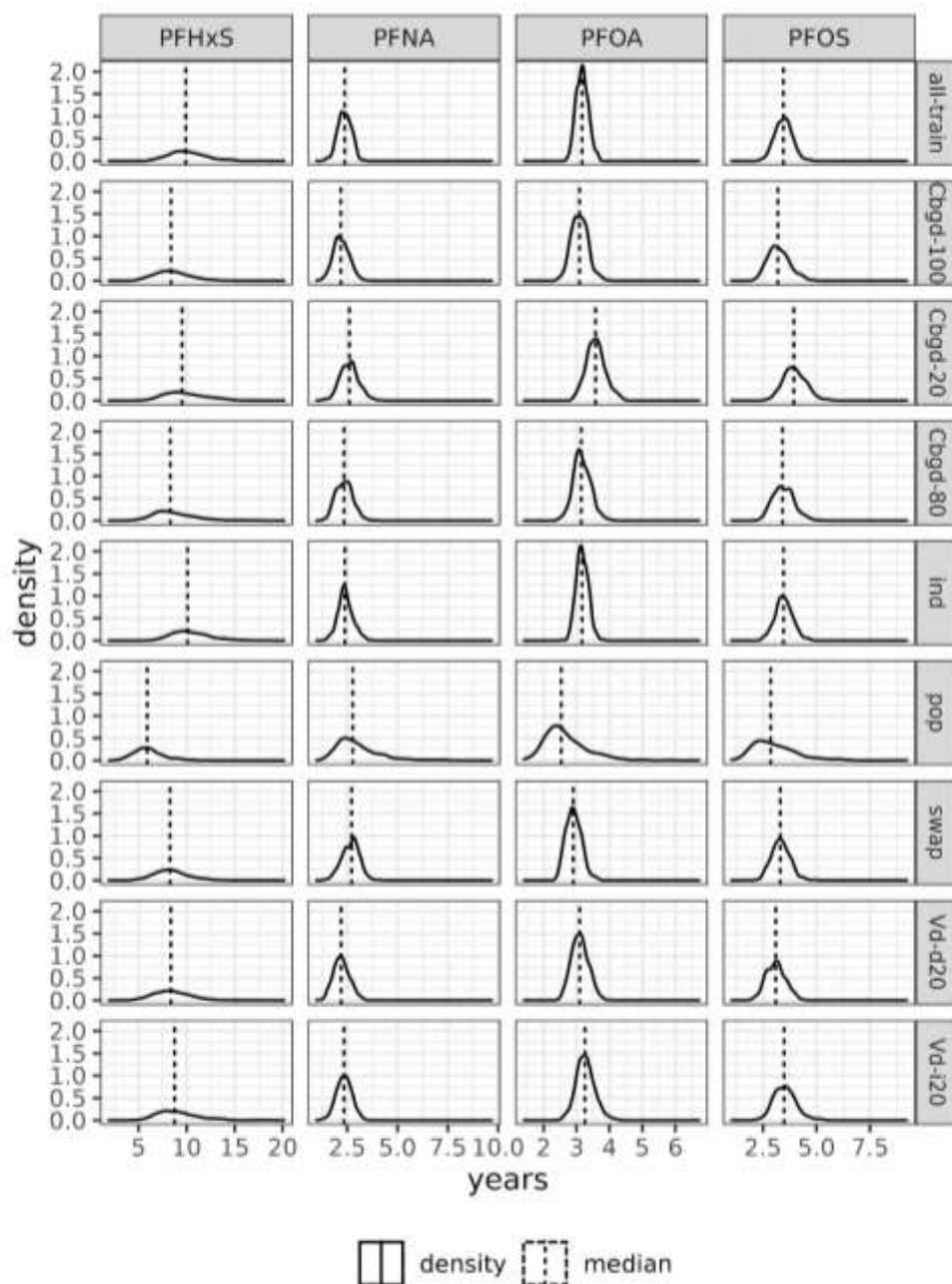


Figure S7. Results of sensitivity analysis showing posterior estimates of geometric standard deviation (GSD) of half-life (dotted line) for each of 9 alternative model definitions (described in methods and in Table S3). all-train = Use all data for training, Cbgd-80 = Primary analysis with background relative source contribution = 20%, Cbgd-100 = Background relative source contribution = 0%, Cbgd-20 = Background relative source contribution = 80%, Vd-d20 = Volume of distribution prior decreased 20%, Vd-i20 = Volume of distribution prior increased 20%, indiv = Use only individual data, pop = Use only population data, swap = Swap training and testing data, PFOA = perfluorooctanoic acid, PFOS = perfluorooctane sulfonic acid, PFNA = perfluorononanoic acid, PFHXS = perfluorohexane sulfonate. More detailed information can be found in the project repository <https://github.com/wachiuphd/2022-Bayes-PFAS-PK>.

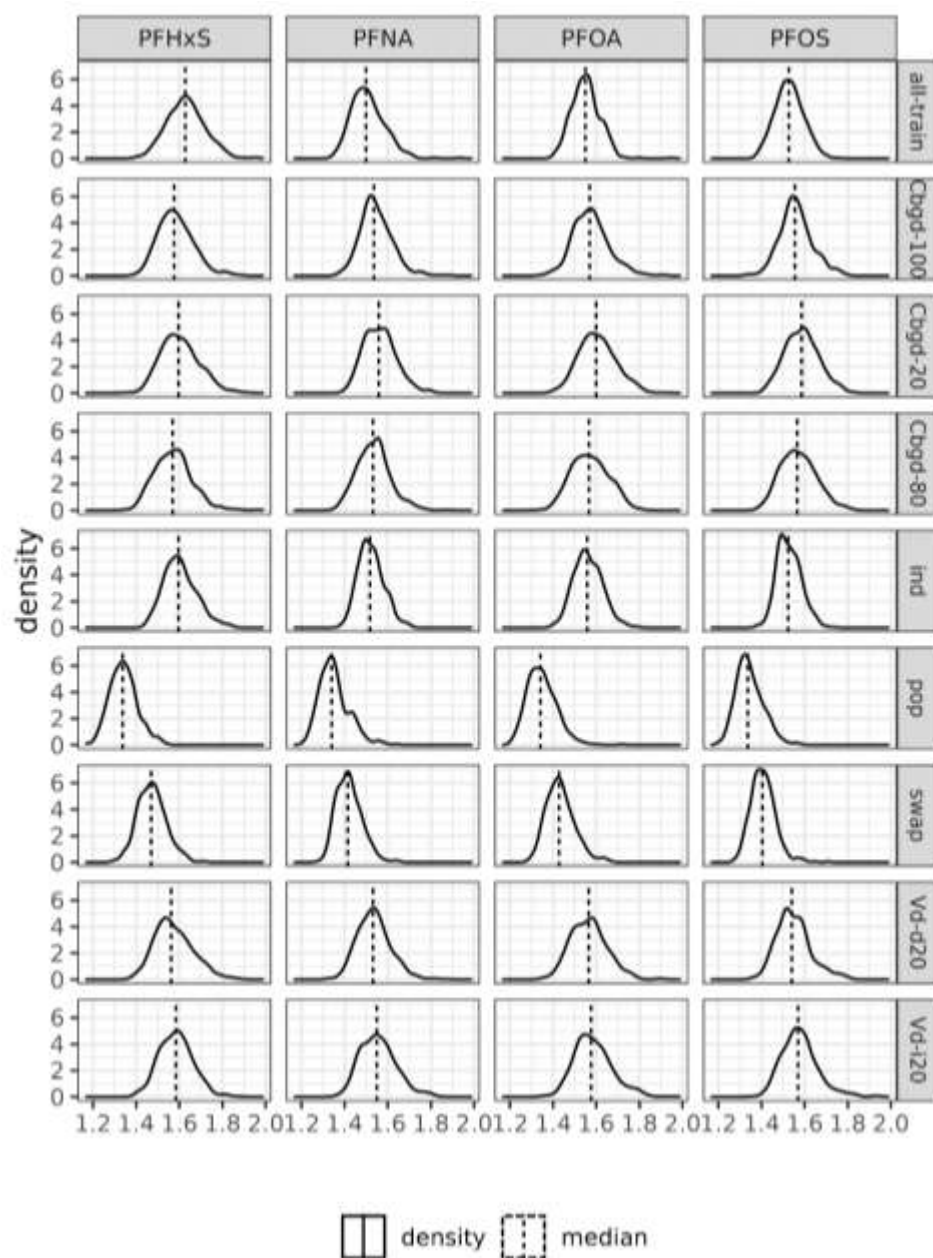


Figure S8. Results of sensitivity analysis showing posterior estimates of geometric mean (GM) of Volume of distribution (Vd) for 9 different model formulations. all-train = Use all data for training, Cbgd-80 = Primary analysis with background relative source contribution = 20%, Cbgd-100 = Background relative source contribution = 0%, Cbgd-20 = Background relative source contribution = 80%, Vd-d20 = Volume of distribution prior decreased 20%, Vd-i20 = Volume of distribution prior increased 20%, indiv = Use only individual data, pop = Use only population data, swap = Swap training and testing data, PFOA = perfluorooctanoic acid, PFOS = perfluorooctane sulfonic acid, PFNA = perfluorononanoic acid, PFHXS = perfluorohexane sulfonate. More detailed information can be found in the project repository <https://github.com/wachiuphd/2022-Bayes-PFAS-PK>.

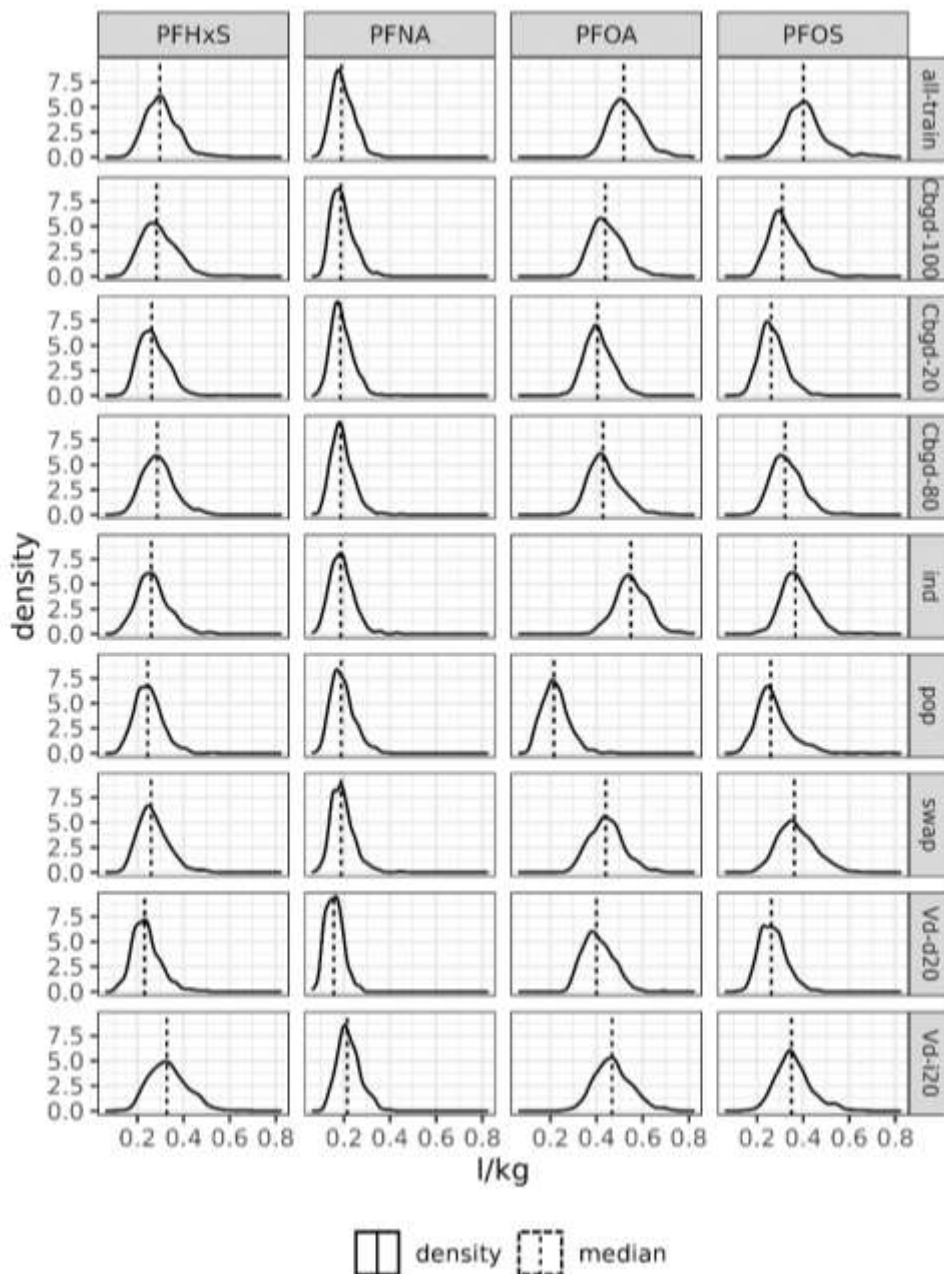
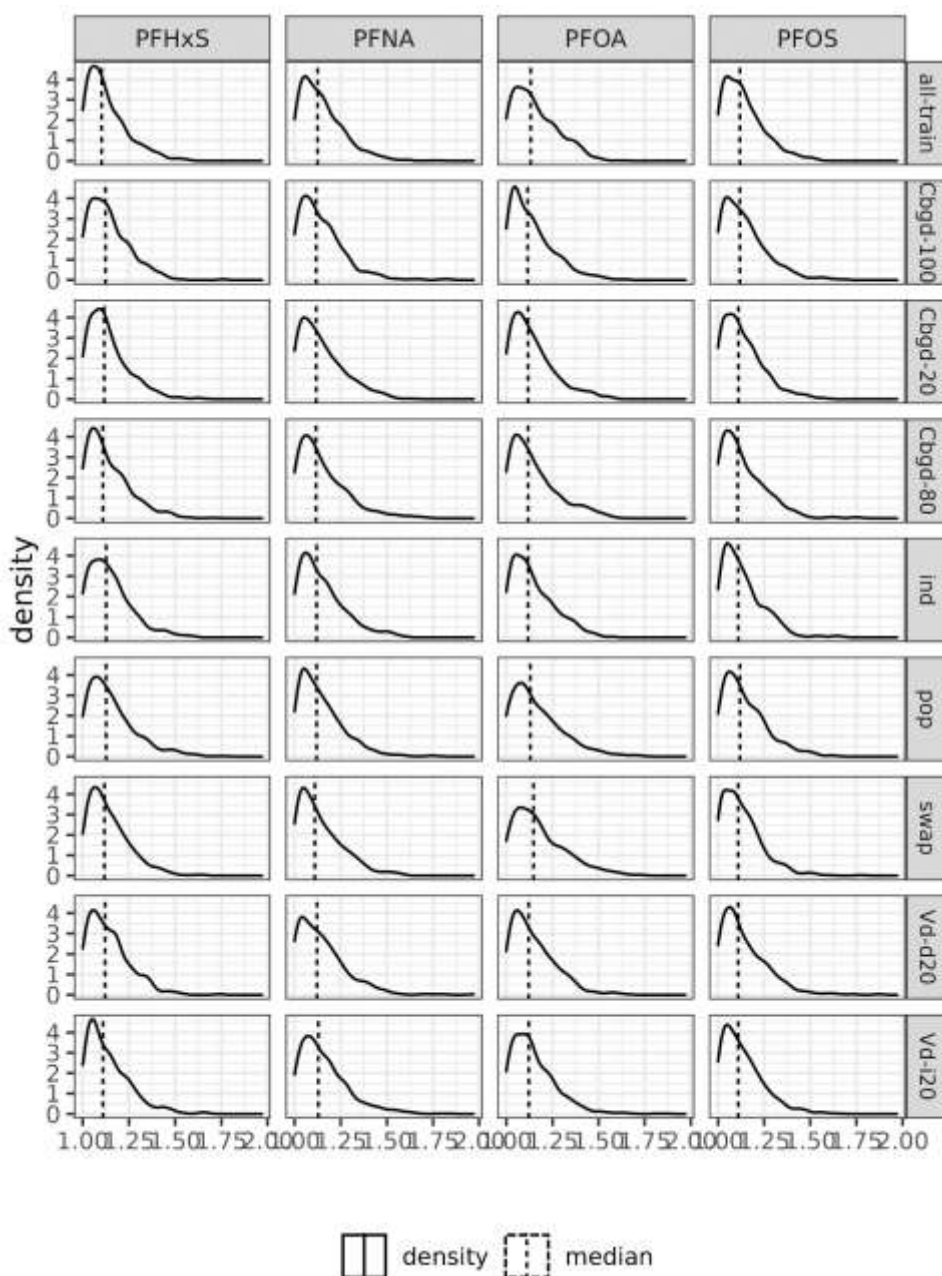


Figure S9. Results of sensitivity analysis for GSD of Vd. Cbgd-80 represents the primary analysis. all-train = Use all data for training, Cbgd-80 = Primary analysis with background relative source contribution = 20%, Cbgd-100 = Background relative source contribution = 0%, Cbgd-20 = Background relative source contribution = 80%, Vd-d20 = Volume of distribution prior decreased 20%, Vd-i20 = Volume of distribution prior increased 20%, indiv = Use only individual data, pop = Use only population data, swap = Swap training and testing data, PFOA = perfluorooctanoic acid, PFOS = perfluorooctane sulfonic acid, PFNA = perfluorononanoic acid, PFHXS = perfluorohexane sulfonate. More detailed information can be found in the project repository <https://github.com/wachiuphd/2022-Bayes-PFAS-PK>.



MCSim Model Code

This extract provides the MCSim code for implementing the model. Key code and tabular outputs are also available in the project-specific repository at:

<https://github.com/wachiuphd/2022-Bayes-PFAS-PK>

```
# Description #####
# Generic single PFAS one-compartment model
#
# Units:
# - time in years (yr) or days (d) [365.25 days/year]
# - volumes in liters (L)
# - PFAS in micrograms (ug)
#
# Population model including error variance
#####
States = {
  Cserum_t # ODE solution for time-varying DWC
};
Inputs = { DWC, # Drinking water concentration (ug/L) (constant)
  DWC_t, # Drinking water concentration (ug/L) (time-varying)
  Cbgd_in_gm, # Background concentration GM (central estimate)
  Cbgd_in_gsd, # Background concentration GSD (central estimate)
  C_0_in_gm, # Initial concentration GM (central estimate)
  C_0_in_gsd}; # Initial concentration GSD (central estimate)

Outputs = {
  # Individual predictions
  Cbgd_Css, Cserum, # Serum concentration (ug/L) including background
  # Population predictions
  M_Cbgd_Css, M_Cserum,
  # Individual parameters
  Cbgd, C_0, k, Vd, DWI_BW_d
};

# Population Mean Parameters
M_ln_Cbgd_sc = 0; # Background serum concentration scaling (e.g., 2.7 ug/L)
M_ln_C_0_sc = 0; # Initial serum concentration scaling (e.g., 1 ug/L)
M_ln_k = -2.5367; # Elimination rate (e.g., 0.07912639 1/yr)
M_ln_Vd = -1.3863; # Distribution volume (e.g., 0.25 L/kg; from MN 2019 DOH
information)
M_ln_DWI_BW_d = -4.395473; # Drinking water intake per kg body weight (e.g., 0.00933
L/(kg d)) See fit_distributions.docx for calculation

# Population SD or Variance Parameters
SD_ln_Cbgd_sc = 1; # Background serum concentration scaling (ug/L)
SD_ln_C_0_sc = 1; # Initial serum concentration scaling (ug/L)
V_ln_k = 0; # Elimination rate (1/yr)
SD_ln_Vd = 0; # Distribution volume (L/kg)
```

```
SD_ln_DWI_BW_d = 0.8880025; # Drinking water intake. See fit_distributions.docx for calculation
```

```
# Individual parameters, log tranformed, z-score
ln_Cbgd = 0; # Background serum concentration (ug/L)
ln_C_0 = 0; # Initial serum concentration (ug/L)
ln_k = 0; # Elimination rate (1/yr)
ln_Vd = 0; # Distribution volume (L/kg)
ln_DWI_BW_d = 0; # Drinking water intake per kg body weight (L/(kg d))
```

```
# Individual Parameters - only can be fixed at individual level (would override sampled values)
```

```
Cbgd = -1; # Background serum concentration (ug/L)
C_0 = -1; # Initial serum concentration (ug/L)
k = -1; # Elimination rate (1/yr)
Vd = -1; # Distribution volume (L/kg)
DWI_BW_d = -1; # Drinking water intake per kg body weight (L/(kg d))
```

```
# Error GSD
```

```
GSD_M_Cserum = 1.2;
GSD_Cserum = 1.2;
GSD_M_Cbgd_Css = 1.2;
GSD_Cbgd_Css = 1.2;
```

```
# DWC below reporting limit
```

```
DWC_belowMRL = 1e-30;
```

```
Initialize {
```

```
  # If specified in simulation file (so >= 0), then use that value
  # Otherwise use sampled value
  Cbgd = (Cbgd < 0) ?
    (Cbgd_in_gm * exp(M_ln_Cbgd_sc + SD_ln_Cbgd_sc * log(Cbgd_in_gsd) * ln_Cbgd )) :
  Cbgd;
  C_0 = (C_0 < 0) ?
    (C_0_in_gm * exp(M_ln_C_0_sc + SD_ln_C_0_sc * log(C_0_in_gsd) * ln_C_0 )) : C_0;
  k = (k < 0) ? exp(M_ln_k + pow(V_ln_k,0.5) * ln_k) : k;
  Vd = (Vd < 0) ? exp(M_ln_Vd + SD_ln_Vd * ln_Vd) : Vd;
  DWI_BW_d = (DWI_BW_d < 0) ?
    exp(M_ln_DWI_BW_d + SD_ln_DWI_BW_d * ln_DWI_BW_d) : DWI_BW_d;
  Cserum_t = C_0;
}
```

```
Dynamics { # only for DWC_t time-varying
```

```
  dt(Cserum_t) = DWI_BW_d * 365.25 * DWC_t / Vd + k*(Cbgd - Cserum_t);
}
```

```
CalcOutputs {
```

```

# Population - exact or approximate arith. mean
# Arith mean of Background
M_Cbgd = Cbgd_in_gm * exp(M_ln_Cbgd_sc + pow(SD_ln_Cbgd_sc *
log(Cbgd_in_gsd),2)/2);
# Arith mean of Css from drinking water
mu_Css = M_ln_DWI_BW_d + log(365.25) + log(DWC) - M_ln_k - M_ln_Vd;
V_Css = pow(SD_ln_DWI_BW_d,2) + V_ln_k + pow(SD_ln_Vd,2);
M_Css = exp(mu_Css + V_Css/2);
# Approximating -kt by normal with matching mean and variance
M_kt = -t*exp(M_ln_k + V_ln_k/2);
V_kt = (exp(V_ln_k)-1)*pow(M_kt,2);
# Arith mean of Background * exp(-kt)
mu_Cbgd_expkt = log(Cbgd_in_gm) + M_ln_Cbgd_sc + M_kt;
V_Cbgd_expkt = pow(SD_ln_Cbgd_sc * log(Cbgd_in_gsd),2) + V_kt;
M_Cbgd_expkt = exp(mu_Cbgd_expkt + V_Cbgd_expkt/2);
# Arith mean of C_0 * exp(-kt)
mu_C_0_expkt = log(C_0_in_gm) + M_ln_C_0_sc + M_kt;
V_C_0_expkt = pow(SD_ln_C_0_sc * log(C_0_in_gsd),2) + V_kt;
M_C_0_expkt = exp(mu_C_0_expkt + V_C_0_expkt/2);
# Arith mean of Css from drinking water * exp(-kt)
mu_Css_expkt = M_ln_DWI_BW_d + log(365.25 * DWC) - M_ln_k - M_ln_Vd + M_kt;
V_Css_expkt = pow(SD_ln_DWI_BW_d,2) + V_ln_k + pow(SD_ln_Vd,2) + V_kt;
M_Css_expkt = exp(mu_Css_expkt + V_Css_expkt/2);
# Arith mean of Cbgd + Css*exp(-kt) from drinking water
M_Cbgd_Css = M_Cbgd + M_Css_expkt;
# Serum concentration as a function of time
M_Cserum = M_Cbgd + M_C_0_expkt - M_Cbgd_expkt + M_Css - M_Css_expkt;

# Individual
Css = DWI_BW_d * 365.25 * DWC / (k * Vd);
# Cbgd + Css*exp(-kt) from drinking water
Cbgd_Css = Cbgd + Css * exp(-k*t);
# time-dependent Cserum with constant DWC
Cserum = Cbgd + (C_0 - Cbgd) * exp(-k*t) + Css * (1 - exp(-k*t));

# Individual Parameters
Cbgd = Cbgd;
C_0 = C_0;
k = k;
Vd = Vd;
DWI_BW_d = DWI_BW_d;
}

End.

```

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